11

Angular Momentum

CHAPTER OUTLINE

11.1 The Vector Product and Torque

11.2 Analysis Model: Nonisolated System (Angular Momentum)

11.3 Angular Momentum of a Rotating Rigid Object

11.4 Analysis Model: Isolated System (Angular Momentum)

11.5 The Motion of Gyroscopes and Tops

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ11.1 Answer (b). Her angular momentum stays constant as I is cut in half and ω doubles. Then $\frac{1}{2}I\omega^2$ doubles.

OQ11.2 The angular momentum of the mouse-turntable system is initially zero, with both at rest. The frictionless axle isolates the mouse-turntable system from outside torques, so its angular momentum must stay constant with the value of zero.

(i) Answer (a). The mouse makes some progress north, or counterclockwise.

(ii) Answer (b). The turntable will rotate clockwise. The turntable rotates in the direction opposite to the motion of the mouse, for the angular momentum of the system to remain zero.

(iii) No. Mechanical energy changes as the mouse converts some chemical into mechanical energy, positive for the motions of both the mouse and the turntable.

(iv) No. Linear momentum is not conserved. The turntable has zero momentum while the mouse has a bit of northward momentum. Initially, momentum is zero; later, when the mouse moves

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north, the fixed axle prevents the turntable from moving south.

(v) Yes. Angular momentum is constant, with the value of zero.

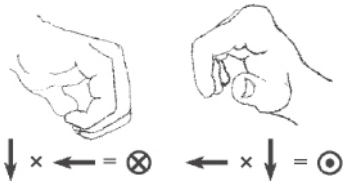
OQ11.3 (i) Answer (a), (ii) Answer (e), (3 m, down) × (2 N, toward you) = 6 N · m, left

OQ11.4 Answer c = e > b = d > a = 0. The unit vectors have magnitude 1, so the magnitude of each cross product is $|1 \cdot 1 \cdot \sin\theta|$ where θ is the angle between the factors. Thus for (a) the magnitude of the cross product is sin 0° = 0. For (b), $|\sin 135°| = 0.707$, (c) sin 90° = 1, (d) sin 45° = 0.707, (e) sin 90° = 1.

OQ11.5 (a) No. (b) No. An axis of rotation must be defined to calculate the torque acting on an object. The moment arm of each force is measured from the axis, so the value of the torque depends on the location of the axis.

OQ11.6 (i) Answer (e). Down–cross–left is away from you: $-\hat{\mathbf{j}} \times \left(-\hat{\mathbf{i}}\right) = -\hat{\mathbf{k}}$, as in the first picture.

(ii) Answer (d). Left–cross–down is toward you: $-\hat{\mathbf{i}} \times \left(-\hat{\mathbf{j}}\right) = \hat{\mathbf{k}}$, as in the second picture.



ANS FIG. OQ11.6

OQ11.7 (i) Answer (a). The angular momentum is constant. The moment of inertia decreases, so the angular speed must increase.

(ii) No. Mechanical energy increases. The ponies must do work to push themselves inward.

(iii) Yes. Momentum stays constant, with the value of zero.

(iv) Yes. Angular momentum is constant with a nonzero value. No outside torque can influence rotation about the vertical axle.

OQ11.8 Answer (d). As long as no net external force, or torque, acts on the system, the linear and angular momentum of the system are constant.

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ11.1 The star is isolated from any outside torques, so its angular momentum is conserved as it changes size. As the radius of the star decreases, its moment of inertia decreases, resulting in its angular speed increasing.

CQ11.2 The suitcase might contain a spinning gyroscope. If the gyroscope is spinning about an axis that is oriented horizontally passing through the bellhop, the force he applies to turn the corner results in a torque that could make the suitcase swing away. If the bellhop turns quickly enough, anything at all could be in the suitcase and need not be rotating. Since the suitcase is massive, it will tend to follow an inertial path. This could be perceived as the suitcase swinging away by the bellhop.

CQ11.3 The long pole has a large moment of inertia about an axis along the rope. An unbalanced torque will then produce only a small angular acceleration of the performer-pole system, to extend the time available for getting back in balance. To keep the center of mass above the rope, the performer can shift the pole left or right, instead of having to bend his body around. The pole sags down at the ends to lower the system's center of gravity.

CQ11.4 (a) Frictional torque arises from kinetic friction between the inside of the roll and the child's fingers. As with all friction, the magnitude of the friction depends on the normal force between the surfaces in contact. As the roll unravels, the weight of the roll decreases, leading to a decrease in the frictional force, and, therefore, a decrease in the torque.

(b) As the radius R of the paper roll shrinks, the roll's angular speed $\omega = \frac{v}{R}$ must increase because the speed v is constant.

(c) If we think of the roll as a uniform disk, then its moment of inertia is $I = \frac{1}{2}MR^2$. But the roll's mass is proportional to its base area πR^2; therefore, the moment of inertia is proportional to R^4. The moment of inertia decreases as the roll shrinks. When the roll is given a sudden jerk, its angular acceleration may not be great enough to set the roll moving in step with the paper, so the paper breaks. The roll is most likely to break when its radius is large, when its moment of inertia is large, than when its radius is small, when its moment of inertia is small.

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CQ11.5 Work done by a torque results in a change in rotational kinetic energy about an axis. Work done by a force results in a change in translational kinetic energy. Work by either has the same units:

$$W = F\Delta x = [\mathrm{N}][\mathrm{m}] = \mathrm{N}\cdot\mathrm{m} = \mathrm{J}$$

$$W = \tau\Delta\theta = [\mathrm{N}\cdot\mathrm{m}][\mathrm{rad}] = \mathrm{N}\cdot\mathrm{m} = \mathrm{J}$$

CQ11.6 Suppose we look at the motorcycle moving to the right. Its drive wheel is turning clockwise. The wheel speeds up when it leaves the ground. No outside torque about its center of mass acts on the airborne cycle, so its angular momentum is conserved. As the drive wheel's clockwise angular momentum increases, the frame of the cycle acquires counterclockwise angular momentum. The cycle's front end moves up and its back end moves down.

CQ11.7 Its angular momentum about that axis is constant in time. You cannot conclude anything about the magnitude of the angular momentum.

CQ11.8 No. The angular momentum about any axis that does not lie along the instantaneous line of motion of the ball is nonzero.

CQ11.9 The Earth is an isolated system, so its angular momentum is conserved when the distribution of its mass changes. When its mass moves away from the axis of rotation, its moment of inertia increases, its angular speed decreases, so its period increases. Most of the mass of Earth would not move, so the effect would be small: we would not have more hours in a day, but more nanoseconds.

CQ11.10 As the cat falls, angular momentum must be conserved. Thus, if the upper half of the body twists in one direction, something must get an equal angular momentum in the opposite direction. Rotating the lower half of the body in the opposite direction satisfies the law of conservation of angular momentum.

CQ11.11 Energy bar charts are useful representations for keeping track of the various types of energy storage in a system: translational and rotational kinetic energy, various types of potential energy, and internal energy. However, there is only *one type* of angular momentum. Therefore, there is no need for bar charts when analyzing a physical situation in terms of angular momentum.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 11.1 The Vector Product and Torque

P11.1 $$\vec{\mathbf{M}}\times\vec{\mathbf{N}}=\begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ 2 & -3 & 1\\ 4 & 5 & -2\end{vmatrix}=\hat{\mathbf{i}}(6-5)-\hat{\mathbf{j}}(-4-4)+\hat{\mathbf{k}}(10+12)=\boxed{\hat{\mathbf{i}}+8.00\hat{\mathbf{j}}+22.0\hat{\mathbf{k}}}$$

P11.2 (a) $$\text{area}=\left|\vec{\mathbf{A}}\times\vec{\mathbf{B}}\right|=AB\sin\theta=(42.0\text{ cm})(23.0\text{ cm})\sin(65.0^\circ-15.0^\circ)$$
$$=\boxed{740\text{ cm}^2}$$

(b) The longer diagonal is equal to the sum of the two vectors.

$$\vec{\mathbf{A}}+\vec{\mathbf{B}}=[(42.0\text{ cm})\cos 15.0^\circ+(23.0\text{ cm})\cos 65.0^\circ]\hat{\mathbf{i}}$$
$$+[(42.0\text{ cm})]\sin 15.0^\circ+(23.0\text{ cm})\sin 65.0^\circ]\hat{\mathbf{j}}$$

$$\vec{\mathbf{A}}+\vec{\mathbf{B}}=(50.3\text{ cm})\hat{\mathbf{i}}+(31.7\text{ cm})\hat{\mathbf{j}}$$

$$\text{length}=\left|\vec{\mathbf{A}}+\vec{\mathbf{B}}\right|=\sqrt{(50.3\text{ cm})^2+(31.7\text{ cm})^2}=\boxed{59.5\text{ cm}}$$

P11.3 We take the cross product of each term of $\vec{\mathbf{A}}$ with each term of $\vec{\mathbf{B}}$, using the cross-product multiplication table for unit vectors. Then we use the identification of the magnitude of the cross product as $AB\sin\theta$ to find θ. We assume the data are known to three significant digits.

(a) We use the definition of the cross product and note that $\hat{\mathbf{i}}\times\hat{\mathbf{i}}=\hat{\mathbf{j}}\times\hat{\mathbf{j}}=0$:

$$\vec{\mathbf{A}}\times\vec{\mathbf{B}}=\left(1\hat{\mathbf{i}}+2\hat{\mathbf{j}}\right)\times\left(2\hat{\mathbf{i}}+3\hat{\mathbf{j}}\right)$$
$$\vec{\mathbf{A}}\times\vec{\mathbf{B}}=2\hat{\mathbf{i}}\times\hat{\mathbf{i}}+3\hat{\mathbf{i}}\times\hat{\mathbf{j}}-4\hat{\mathbf{j}}\times\hat{\mathbf{i}}+6\hat{\mathbf{j}}\times\hat{\mathbf{j}}$$
$$=0+3\hat{\mathbf{k}}-4\left(-\hat{\mathbf{k}}\right)+0=\boxed{7.00\hat{\mathbf{k}}}$$

(b) Since $\left|\vec{\mathbf{A}}\times\vec{\mathbf{B}}\right|=AB\sin\theta$, we have

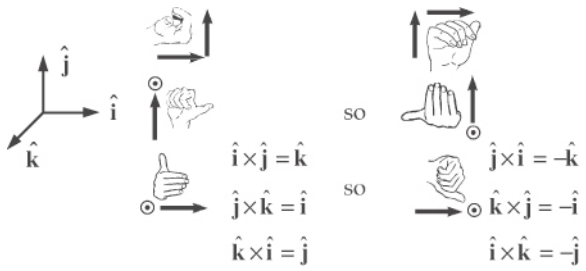
$$\theta=\sin^{-1}\left(\frac{\left|\vec{\mathbf{A}}\times\vec{\mathbf{B}}\right|}{AB}\right)=\sin^{-1}\left(\frac{7}{\sqrt{1^2+2^2}\sqrt{2^2+3^2}}\right)=\boxed{60.3^\circ}$$

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P11.4 $\left|\hat{\mathbf{i}} \times \hat{\mathbf{i}}\right| = 1 \cdot 1 \cdot \sin 0° = 0$

$\hat{\mathbf{j}} \times \hat{\mathbf{j}}$ and $\hat{\mathbf{k}} \times \hat{\mathbf{k}}$ are zero similarly since the vectors being multiplied are parallel.

$\left|\hat{\mathbf{i}} \times \hat{\mathbf{j}}\right| = 1 \cdot 1 \cdot \sin 90° = 1$



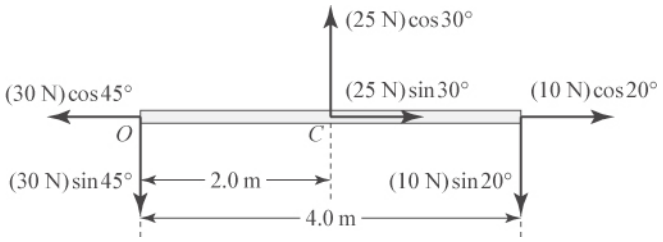
$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$ $\quad$ $\hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}$

$\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$ $\quad$ $\hat{\mathbf{k}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}$

$\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$ $\quad$ $\hat{\mathbf{i}} \times \hat{\mathbf{k}} = -\hat{\mathbf{j}}$

ANS. FIG. P11.4

P11.5 We first resolve all of the forces shown in Figure P11.5 into components parallel to and perpendicular to the beam as shown in ANS. FIG. P11.5.



ANS. FIG. P11.5

(a) The torque about an axis through point *O* is given by

$$\tau_O = +\left[(25\text{ N})\cos 30°\right](2.0\text{ m}) - \left[(10\text{ N})\sin 20°\right](4.0\text{ m}) = \boxed{+30\text{ N}\cdot\text{m}}$$

or $\tau_0 = \boxed{30\text{ N}\cdot\text{m counterclockwise}}$

(b) The torque about an axis through point *C* is given by

$$\tau_C = +\left[(30\text{ N})\sin 45°\right](2.0\text{ m}) - \left[(10\text{ N})\sin 20°\right](2.0\text{ m}) = \boxed{+36\text{ N}\cdot\text{m}}$$

or $\tau_C = \boxed{36\text{ N}\cdot\text{m counterclockwise}}$

P11.6 $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = -3.00(6.00) + 7.00(-10.0) + (-4.00)(9.00)$

$= -124$

$$AB = \sqrt{(-3.00)^2 + (7.00)^2 + (-4.00)^2} \cdot \sqrt{(6.00)^2 + (-10.0)^2 + (9.00)^2}$$

$= 127$

(a) $\cos^{-1}\left(\dfrac{\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}}{AB}\right) = \cos^{-1}(-0.979) = \boxed{168°}$

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(b) $$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -3.00 & 7.00 & -4.00 \\ 6.00 & -10.0 & 9.00 \end{vmatrix} = 23.0\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}} - 12.0\hat{\mathbf{k}}$$

$$\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = \sqrt{(23.0)^2 + (3.00)^2 + (-12.0)^2} = 26.1$$

$$\sin^{-1}\left(\frac{\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|}{AB}\right) = \sin^{-1}(0.206) = \boxed{11.9^\circ} \text{ or } 168^\circ$$

(c) Only the first method gives the angle between the vectors unambiguously because $\sin(180^\circ - \theta) = \sin\theta$ but $\cos(180^\circ - \theta) = -\cos\theta$; in other words, the vectors can only be at most 180° apart and using the second method cannot distinguish θ from $180^\circ - \theta$.

P11.7 We are given the condition $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = \vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$.

This says that $AB\sin\theta = AB\cos\theta$

so $\tan\theta = 1$

$\theta = \boxed{45.0^\circ}$ satisfies this condition.

P11.8 (a) The torque acting on the particle about the origin is

$$\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 4 & 6 & 0 \\ 3 & 2 & 0 \end{vmatrix} = \hat{\mathbf{i}}(0-0) - \hat{\mathbf{j}}(0-0) + \hat{\mathbf{k}}(8-18)$$

$$= \boxed{(-10.0\ \text{N}\cdot\text{m})\hat{\mathbf{k}}}$$

(b) Yes. The point or axis must be on the other side of the line of action of the force, and half as far from this line along which the force acts. Then the lever arm of the force about this new axis will be half as large and the force will produce counter-clockwise instead of clockwise torque.

(c) Yes. There are infinitely many such points, along a line that passes through the point described in (b) and parallel the line of action of the force.

(d) Yes, at the intersection of the line described in (c) and the y axis.

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(e) No, because there is only one point of intersection of the line described in (d) with the y axis.

(f) Let $(0, y)$ represent the coordinates of the special axis of rotation located on the y axis of Cartesian coordinates. Then the displacement from this point to the particle feeling the force is $\vec{\mathbf{r}}_{\text{new}} = 4\hat{\mathbf{i}} + (6-y)\hat{\mathbf{j}}$ in meters. The torque of the force about this new axis is

$$\vec{\boldsymbol{\tau}}_{\text{new}} = \vec{\mathbf{r}}_{\text{new}} \times \vec{\mathbf{F}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 4 & 6-y & 0 \\ 3 & 2 & 0 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(0-0) - \hat{\mathbf{j}}(0-0) + \hat{\mathbf{k}}(8-18+3y)$$

$$= (+5\ \text{N}\cdot\text{m})\hat{\mathbf{k}}$$

Then,

$$8-18+3y=5 \quad \rightarrow \quad 3y=15 \quad \rightarrow \quad y=5$$

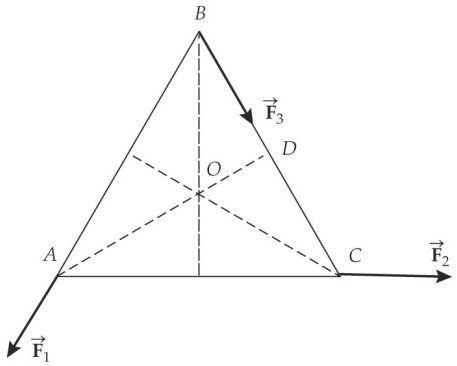
The position vector of the new axis is $5.00\hat{\mathbf{j}}\ \text{m}$.

P11.9 (a) The lever arms of the forces about O are all the same, equal to length OD, L.

If $\vec{\mathbf{F}}_3$ has a magnitude $\left|\vec{\mathbf{F}}_3\right| = \left|\vec{\mathbf{F}}_1\right| + \left|\vec{\mathbf{F}}_2\right|$, the net torque is zero:

$$\sum \tau = F_1 L + F_2 L - F_3 L = F_1 L + F_2 L - (F_1 + F_2) L = 0$$

(b) The torque produced by $\vec{\mathbf{F}}_3$ depends on the perpendicular distance OD, therefore translating the point of application of $\vec{\mathbf{F}}_3$ to any other point along BC will not change the net torque.



ANS. FIG. P11.9

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P11.10 (a) No.

(b) The cross-product vector must be perpendicular to both of the factors, so its dot product with either factor must be zero. To check:

$$\left(2\hat{\mathbf{i}}-3\hat{\mathbf{j}}+4\hat{\mathbf{k}}\right)\cdot\left(4\hat{\mathbf{i}}+3\hat{\mathbf{j}}-\hat{\mathbf{k}}\right)=\left(\hat{\mathbf{i}}\cdot\hat{\mathbf{i}}\right)8+-9\left(\hat{\mathbf{j}}\cdot\hat{\mathbf{j}}\right)-4\left(\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}\right)$$
$$=8-9-4=-5$$

The answer is not zero.

No. The cross product could not work out that way.

Section 11.2 Analysis Model: Nonisolated System (Angular Momentum)

P11.11 Taking the geometric center of the compound object to be the pivot, the angular speed and the moment of inertia are

$$\omega = v/r = (5.00\text{ m/s})/0.500\text{ m} = 10.0\text{ rad/s}$$

and

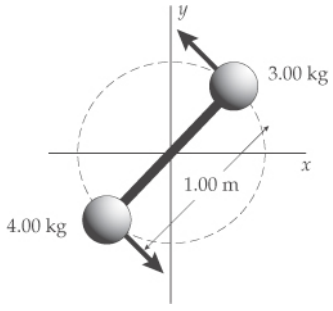
$$I=\sum mr^2=\left(4.00\text{ kg}\right)(0.500\text{ m})^2+\left(3.00\text{ kg}\right)(0.500\text{ m})^2$$
$$=1.75\text{ kg}\cdot\text{m}^2$$

By the right-hand rule, we find that the angular velocity is directed out of the plane. So the object's angular momentum, with magnitude

$$L=I\omega=\left(1.75\text{ kg}\cdot\text{m}^2\right)(10.0\text{ rad/s})$$

is the vector

$$\vec{\mathbf{L}}=\left(17.5\text{ kg}\cdot\text{m}^2/\text{s}\right)\hat{\mathbf{k}}$$



ANS. FIG. P11.11

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P11.12 We use $\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$:

$$\vec{\mathbf{L}} = \left(1.50\hat{\mathbf{i}} + 2.20\hat{\mathbf{j}}\right) \text{ m} \times (1.50 \text{ kg})\left(4.20\hat{\mathbf{i}} - 3.60\hat{\mathbf{j}}\right) \text{ m/s}$$

$$\vec{\mathbf{L}} = \left(-8.10\hat{\mathbf{k}} - 13.9\hat{\mathbf{k}}\right) \text{ kg} \cdot \text{m}^2/\text{s} = \boxed{\left(-22.0 \text{ kg} \cdot \text{m}^2/\text{s}\right)\hat{\mathbf{k}}}$$

P11.13 We use $\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$:

$$\vec{\mathbf{L}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ x & y & 0 \\ mv_x & mv_y & 0 \end{vmatrix} = \hat{\mathbf{i}}(0-0) - \hat{\mathbf{j}}(0-0) + \hat{\mathbf{k}}\left(mxv_y - myv_x\right)$$

$$\vec{\mathbf{L}} = \boxed{m\left(xv_y - yv_x\right)\hat{\mathbf{k}}}$$

P11.14 Whether we think of the Earth's surface as curved or flat, we interpret the problem to mean that the plane's line of flight extended is precisely tangent to the mountain at its peak, and nearly parallel to the wheat field. Let the positive x direction be eastward, positive y be northward, and positive z be vertically upward.

(a) $\vec{\mathbf{r}} = (4.30 \text{ km})\hat{\mathbf{k}} = \left(4.30 \times 10^3 \text{ m}\right)\hat{\mathbf{k}}$

$$\vec{\mathbf{p}} = m\vec{\mathbf{v}} = \left(12\,000 \text{ kg}\right)\left(-175\hat{\mathbf{i}} \text{ m/s}\right) = -2.10 \times 10^6 \hat{\mathbf{i}} \text{ kg} \cdot \text{m/s}$$

$$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = \left(4.30 \times 10^3 \hat{\mathbf{k}} \text{ m}\right) \times \left(-2.10 \times 10^6 \hat{\mathbf{i}} \text{ kg} \cdot \text{m/s}\right)$$

$$= \boxed{\left(-9.03 \times 10^9 \text{ kg} \cdot \text{m}^2/\text{s}\right)\hat{\mathbf{j}}}$$

(b) No. $L = |\vec{\mathbf{r}}||\vec{\mathbf{p}}| \sin\theta = mv(r\sin\theta)$, and $r\sin\theta$ is the altitude of the plane. Therefore, L = *constant* as the plane moves in level flight with constant velocity.

(c) Zero. The position vector from Pike's Peak to the plane is anti-parallel to the velocity of the plane. That is, it is directed along the same line and opposite in direction. Thus, $L = mvr \sin 180° = 0$.

P11.15 (a) Zero because $\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$ and $\vec{\mathbf{r}} = 0$.

(b) At the highest point of the trajectory,

$$x = \frac{1}{2}R = \frac{v_i^2 \sin 2\theta}{2g} \text{ and}$$

$$y = h_{\max} = \frac{\left(v_i \sin\theta\right)^2}{2g}$$

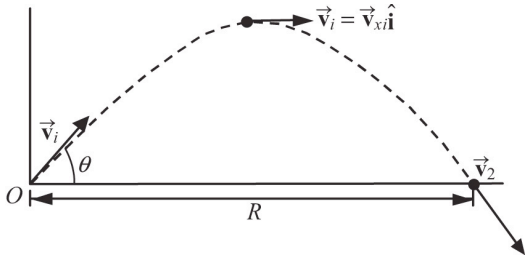
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The angular momentum is then

$$\vec{\mathbf{L}}_1 = \vec{\mathbf{r}}_1 \times m\vec{\mathbf{v}}_1$$

$$= \left[\frac{v_i^2 \sin 2\theta}{2g}\hat{\mathbf{i}} + \frac{(v_i \sin\theta)^2}{2g}\hat{\mathbf{j}} \right] \times m v_{xi}\hat{\mathbf{i}}$$

$$= \boxed{\frac{-mv_i^3 \sin\theta^2 \cos\theta}{2g}\hat{\mathbf{k}}}$$



ANS. FIG. P11.15

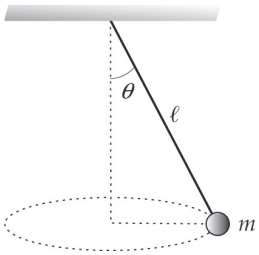
(c) $\vec{\mathbf{L}}_2 = R\hat{\mathbf{i}} \times m\vec{\mathbf{v}}_2$, where $R = \dfrac{v_i^2 \sin 2\theta}{g} = \dfrac{v_i^2 (2\sin\theta\cos\theta)}{g}$

$$= mR\hat{\mathbf{i}} \times \left(v_i \cos\theta\, \hat{\mathbf{i}} - v_i \sin\theta\, \hat{\mathbf{j}} \right)$$

$$= -mRv_i \sin\theta\, \hat{\mathbf{k}} = \boxed{\frac{-2mv_i^3 \sin\theta \sin\theta}{g}\hat{\mathbf{k}}}$$

(d) The downward force of gravity exerts a torque in the $-z$ direction.

P11.16 We start with the particle under a net force model in the x and y directions:



ANS. FIG. P11.16

$$\sum F_x = ma_x: \qquad T\sin\theta = \frac{mv^2}{r}$$

$$\sum F_y = ma_y: \qquad T\cos\theta = mg$$

So $$\frac{\sin\theta}{\cos\theta} = \frac{v^2}{rg} \quad \text{and} \quad v = \sqrt{rg\frac{\sin\theta}{\cos\theta}}$$

then $$L = rmv\sin 90.0° = rm\sqrt{rg\frac{\sin\theta}{\cos\theta}} = \sqrt{m^2 g r^3 \frac{\sin\theta}{\cos\theta}}$$

and since $r = \ell\sin\theta$,

$$L = \boxed{\sqrt{m^2 g \ell^3 \frac{\sin^4\theta}{\cos\theta}}}$$

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P11.17 The angular displacement of the particle around the circle is $\theta = \omega t = \frac{vt}{R}$.

The vector from the center of the circle to the mass is then

$R\cos\theta\,\hat{\mathbf{i}} + R\sin\theta\,\hat{\mathbf{j}}$, where R is measured from the $+x$ axis.

The vector from point P to the mass is

$$\vec{\mathbf{r}} = R\hat{\mathbf{i}} + R\cos\theta\,\hat{\mathbf{i}} + R\sin\theta\,\hat{\mathbf{j}}$$

$$\vec{\mathbf{r}} = R\left[\left(1+\cos\left(\frac{vt}{R}\right)\right)\hat{\mathbf{i}} + \sin\left(\frac{vt}{R}\right)\hat{\mathbf{j}}\right]$$

The velocity is

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt} = -v\sin\left(\frac{vt}{R}\right)\hat{\mathbf{i}} + v\cos\left(\frac{vt}{R}\right)\hat{\mathbf{j}}$$

So

$$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times m\vec{\mathbf{v}}$$

$$\vec{\mathbf{L}} = mvR\left[(1+\cos\omega t)\hat{\mathbf{i}} + \sin\omega t\hat{\mathbf{j}}\right] \times \left[-\sin\omega t\hat{\mathbf{i}} + \cos\omega t\hat{\mathbf{j}}\right]$$

$$\vec{\mathbf{L}} = \boxed{mvR\left[\cos\left(\frac{vt}{R}\right)+1\right]\hat{\mathbf{k}}}$$

P11.18 (a) The net torque on the counterweight-cord-spool system is

$$\tau = \left|\vec{\mathbf{r}} \times \vec{\mathbf{F}}\right| = Rmg\sin\theta$$

$$\tau = 8.00\times10^{-2}\ \text{m}\left(4.00\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)\sin 90.0° = \boxed{3.14\ \text{N}\cdot\text{m}}$$

(b) $$L = \sum_i \left|\vec{\mathbf{r}} \times m_i\vec{\mathbf{v}}_i\right| = Rmv + RMv = R(m+M)v$$

$$L = (0.080\ 0\ \text{m})\left(4.00\ \text{kg} + 2.00\ \text{kg}\right)v = \boxed{(0.480\ \text{kg}\cdot\text{m})v}$$

(c) $$\tau = \frac{dL}{dt} = \left(0.480\ \text{kg}\cdot\text{m}\right)a \quad \rightarrow \quad a = \frac{3.14\ \text{N}\cdot\text{m}}{0.480\ \text{kg}\cdot\text{m}} = \boxed{6.53\ \text{m/s}^2}$$

P11.19 Differentiating $\vec{\mathbf{r}} = \left(6.00\hat{\mathbf{i}} + 5.00t\hat{\mathbf{j}}\ \text{m}\right)$ with respect to time gives

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt} = 5.00\hat{\mathbf{j}}\ \text{m/s}$$

so $$\vec{\mathbf{p}} = m\vec{\mathbf{v}} = \left(2.00\ \text{kg}\right)\left(5.00\hat{\mathbf{j}}\ \text{m/s}\right) = 10.0\hat{\mathbf{j}}\ \text{kg}\cdot\text{m/s}$$

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and $$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 6.00 & 5.00t & 0 \\ 0 & 10.0 & 0 \end{vmatrix} = \boxed{\left(60.0 \text{ kg} \cdot \text{m}^2/\text{s}\right)\hat{\mathbf{k}}}$$

P11.20 (a) $$\int_0^{\vec{r}} d\vec{\mathbf{r}} = \int_0^t \vec{\mathbf{v}} dt = \vec{\mathbf{r}} - 0 = \int_0^t (6t^2\hat{\mathbf{i}} + 2t\hat{\mathbf{j}})dt = \vec{\mathbf{r}} = \left(6t^3/3\right)\hat{\mathbf{i}} + \left(2t^2/2\right)\hat{\mathbf{j}}$$

$$= \boxed{2t^3\hat{\mathbf{i}} + t^2\hat{\mathbf{j}}} \text{ in meters, where } t \text{ is in seconds.}$$

(b) The particle starts from rest at the origin, starts moving into the first quadrant, and gains speed faster and faster while turning to move more and more nearly parallel to the x axis.

(c) $$\vec{\mathbf{a}} = (d\vec{\mathbf{v}}/dt) = (d/dt)(6t^2\,\hat{\mathbf{i}} + 2t\,\hat{\mathbf{j}}) = \boxed{(12t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}})\ \text{m/s}^2}$$

(d) $$\vec{\mathrm{F}} = m\vec{\mathbf{a}} = (5\text{ kg})(12t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}})\ \text{m/s}^2 = \boxed{(60t\,\hat{\mathbf{i}} + 10\,\hat{\mathbf{j}})\ \text{N}}$$

(e) $$\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}} = (2t^3\hat{\mathbf{i}} + t^2\hat{\mathbf{j}}) \times (60t\hat{\mathbf{i}} + 10\hat{\mathbf{j}}) = 20t^3\hat{\mathbf{k}} - 60t^3\hat{\mathbf{k}}$$

$$= \boxed{-40t^3\hat{\mathbf{k}}\ \text{N} \cdot \text{m}}$$

(f) $$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times m\vec{\mathbf{v}} = (5\text{ kg})(2t^3\hat{\mathbf{i}} + t^2\hat{\mathbf{j}}) \times (6t^2\hat{\mathbf{i}} + 2t\hat{\mathbf{j}}) = 5(4t^4\hat{\mathbf{k}} - 6t^4\hat{\mathbf{k}})$$

$$= \boxed{-10t^4\hat{\mathbf{k}}\ \text{kg} \cdot \text{m}^2/\text{s}}$$

(g) $$K = \frac{1}{2}m\vec{\mathbf{v}} \cdot \vec{\mathbf{v}} = \frac{1}{2}(5\text{ kg})(6t^2\hat{\mathbf{i}} + 2t\hat{\mathbf{j}}) \cdot (6t^2\hat{\mathbf{i}} + 2t\hat{\mathbf{j}}) = (2.5)(36t^4 + 4t^2)$$

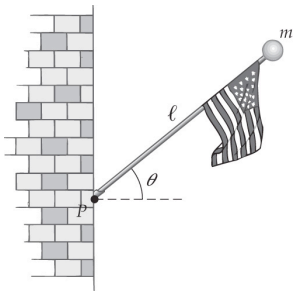
$$= \boxed{(90t^4 + 10t^2)\ \text{J}}$$

(h) $$P = (d/dt)(90t^4 + 10t^2)\ \text{J} = \boxed{(360t^3 + 20t)\ \text{W}},$$ all where t is in seconds.

P11.21 (a) The vector from P to the falling ball is

$$\vec{\mathbf{r}} = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2}\vec{\mathbf{a}}t^2$$

$$\vec{\mathbf{r}} = \left(\ell\cos\theta\,\hat{\mathbf{i}} + \ell\sin\theta\,\hat{\mathbf{j}}\right) + 0 - \left(\frac{1}{2}gt^2\right)\hat{\mathbf{j}}$$



ANS. FIG. P11.21

The velocity of the ball is

$$\vec{\mathbf{v}} = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t = 0 - gt\hat{\mathbf{j}}$$

So $$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times m\vec{\mathbf{v}}$$

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$$\vec{\mathbf{L}} = m\left[\left(\ell\cos\theta\,\hat{\mathbf{i}} + \ell\sin\theta\,\hat{\mathbf{j}}\right) + 0 - \left(\frac{1}{2}gt^2\right)\hat{\mathbf{j}}\right] \times \left(-gt\hat{\mathbf{j}}\right)$$

$$\vec{\mathbf{L}} = \boxed{-mg\ell\, t\cos\theta\,\hat{\mathbf{k}}}$$

(b) The Earth exerts a gravitational torque on the projectile in the negative z direction.

(c) Differentiating with respect to time, we have $\boxed{-mg\ell\cos\theta\,\hat{\mathbf{k}}}$ for the rate of change of angular momentum, which is also the torque due to the gravitational force on the ball.

Section 11.3 Angular Momentum of a Rotating Rigid Object

P11.22 The moment of inertia of the sphere about an axis through its center is

$$I = \frac{2}{5}MR^2 = \frac{2}{5}(15.0\text{ kg})(0.500\text{ m})^2 = 1.50\text{ kg}\cdot\text{m}^2$$

Therefore, the magnitude of the angular momentum is

$$L = I\omega = (1.50\text{ kg}\cdot\text{m}^2)(3.00\text{ rad/s}) = 4.50\text{ kg}\cdot\text{m}^2/\text{s}$$

Since the sphere rotates counterclockwise about the vertical axis, the angular momentum vector is directed upward in the $+z$ direction.

Thus,

$$\boxed{\vec{\mathbf{L}} = (4.50\text{ kg}\cdot\text{m}^2/\text{s})\hat{\mathbf{k}}}$$

P11.23 The total angular momentum about the center point is given by

$$L = I_h\omega_h + I_m\omega_m$$

For the hour hand: $$I_h = \frac{m_hL_h^2}{3} = \frac{60.0\text{ kg}(2.70\text{ m})^2}{3} = 146\text{ kg}\cdot\text{m}^2$$

For the minute hand: $$I_m = \frac{m_mL_m^2}{3} = \frac{100\text{ kg}(4.50\text{ m})^2}{3} = 675\text{ kg}\cdot\text{m}^2$$

In addition, $$\omega_h = \frac{2\pi\text{ rad}}{12\text{ h}}\left(\frac{1\text{ h}}{3\,600\text{ s}}\right) = 1.45\times10^{-4}\text{ rad/s}$$

while $$\omega_m = \frac{2\pi\text{ rad}}{1\text{ h}}\left(\frac{1\text{ h}}{3600\text{ s}}\right) = 1.75\times10^{-3}\text{ rad/s}$$

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Thus, $L = (146\ \text{kg}\cdot\text{m}^2)(1.45\times10^{-4}\ \text{rad/s}) + (675\ \text{kg}\cdot\text{m}^2)(1.75\times10^{-3}\ \text{rad/s})$

or $\boxed{L = 1.20\ \text{kg}\cdot\text{m}^2/\text{s}}$. The hands turn clockwise, so their vector angular momentum is $\boxed{\text{perpendicularly into the clock face.}}$

P11.24 We begin with

$$K = \frac{1}{2}I\omega^2$$

And multiply the right-hand side by $\frac{I}{I}$:

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}\frac{I^2\omega^2}{I}$$

Substituting $L = I\omega$ then gives

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}\frac{I^2\omega^2}{I} = \boxed{\frac{L^2}{2I}}$$

P11.25 (a) For an axis of rotation passing through the center of mass, the magnitude of the angular momentum is given by

$$L = I\omega = \left(\frac{1}{2}MR^2\right)\omega = \frac{1}{2}(3.00\ \text{kg})(0.200\ \text{m})^2(6.00\ \text{rad/s})$$
$$= \boxed{0.360\ \text{kg}\cdot\text{m}^2/\text{s}}$$

(b) For a point midway between the center and the rim, we use the parallel-axis theorem to find the moment of inertia about this point. Then,

$$L = I\omega = \left[\frac{1}{2}MR^2 + M\left(\frac{R}{2}\right)^2\right]\omega$$
$$= \frac{3}{4}(3.00\ \text{kg})(0.200\ \text{m})^2(6.00\ \text{rad/s}) = \boxed{0.540\ \text{kg}\cdot\text{m}^2/\text{s}}$$

P11.26 (a) Modeling the Earth as a sphere, we first calculate its moment of inertia about its rotation axis.

$$I = \frac{2}{5}MR^2 = \frac{2}{5}(5.98\times10^{24}\ \text{kg})(6.37\times10^{6}\ \text{m})^2$$
$$= 9.71\times10^{37}\ \text{kg}\cdot\text{m}^2$$

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Completing one rotation in one day, Earth's rotational angular speed is

$$\omega = \frac{1 \text{ rev}}{24 \text{ h}} = \frac{2\pi \text{ rad}}{86\ 400 \text{ s}} = 7.27 \times 10^{5} \text{ s}^{-1}$$

the rotational angular momentum of the Earth is then

$$L = I\omega = \left(9.71 \times 10^{37} \text{ kg} \cdot \text{m}^2\right)\left(7.27 \times 10^{5} \text{ s}^{-1}\right)$$

$$= \boxed{7.06 \times 10^{33} \text{ kg} \cdot \text{m}^2/\text{s}}$$

The Earth turns toward the east, counterclockwise as seen from above north, so the vector angular momentum points north along the Earth's axis, $\boxed{\text{towards the north celestial pole}}$ or nearly toward the star Polaris.

(b) In this case, we model the Earth as a particle, with moment of inertia

$$I = MR^2 = \left(5.98 \times 10^{24} \text{ kg}\right)\left(1.496 \times 10^{11} \text{ m}\right)^2$$

$$= 1.34 \times 10^{47} \text{ kg} \cdot \text{m}^2$$

Completing one orbit in one year, Earth's orbital angular speed is

$$\omega = \frac{1 \text{ rev}}{365.25 \text{ d}} = \frac{2\pi \text{ rad}}{(365.25 \text{ d})(86\ 400 \text{ s/d})} = 1.99 \times 10^{-7} \text{ s}^{-1}$$

the angular momentum of the Earth is then

$$L = I\omega = \left(1.34 \times 10^{47} \text{ kg} \cdot \text{m}^2\right)\left(1.99 \times 10^{-7} \text{ s}^{-1}\right)$$

$$= \boxed{2.66 \times 10^{40} \text{ kg} \cdot \text{m}^2/\text{s}}$$

The Earth plods around the Sun, counterclockwise as seen from above north, so the vector angular momentum points north perpendicular to the plane of the ecliptic,
$\boxed{\text{toward the north ecliptic pole}}$ or 23.5° away from Polaris, toward the center of the circle that the north celestial pole moves in as the equinoxes precess. The north ecliptic pole is in the constellation Draco.

(c) $\boxed{\text{The periods differ only by a factor of 365 (365 days for orbital motion to 1 day for rotation). Because of the huge distance from the Earth to the Sun, however, the moment of inertia of the Earth around the Sun is six orders of magnitude larger than that of the Earth about its axis.}}$

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P11.27 Defining the distance from the pivot to the particle as d, we first find the rotational inertia of the system for each case, from the information $M = 0.100$ kg, $m = 0.400$ kg, and $D = 1.00$ m.

(a) For the meterstick rotated about its center, $I_m = \frac{1}{12}MD^2$.

For the additional particle, $I_w = md^2 = m\left(\frac{1}{2}D^2\right)$.

Together, $I = I_m + I_w = \frac{1}{12}MD^2 + \frac{1}{4}mD^2$, or

$$I = \frac{(0.100\text{ kg})(1.00\text{ m})^2}{12} + \frac{(0.400\text{ kg})(1.00\text{ m})^2}{4} = 0.108\text{ kg}\cdot\text{m}^2$$

And the angular momentum is

$$L = I\omega = \left(0.108\text{ kg}\cdot\text{m}^2\right)(4.00\text{ rad/s}) = \boxed{0.433\text{ kg}\cdot\text{m}^2/\text{s}}$$

(b) For a stick rotated about a point at one end,

$$I_m = \frac{1}{3}mD^2 = \frac{1}{3}(0.100\text{ kg})(1.00\text{ m})^2 = 0.033\ 3\text{ kg}\cdot\text{m}^2$$

For a point mass, $I_w = mD^2 = (0.400\text{ kg})(1.00\text{ m})^2 = 0.400\text{ kg}\cdot\text{m}^2$

so together they have rotational inertia

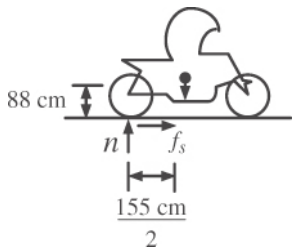
$$I = I_m + I_w = 0.433\text{ kg}\cdot\text{m}^2$$

and angular momentum

$$L = I\omega = \left(0.433\text{ kg}\cdot\text{m}^2\right)(4.00\text{ rad/s}) = \boxed{1.73\text{ kg}\cdot\text{m}^2/\text{s}}$$

P11.28 We assume that the normal force $n = 0$ on the front wheel. On the bicycle,

$$\sum F_x = ma_x: \qquad +f_s = ma_x$$
$$\sum F_y = ma_y: \qquad +n - F_g = 0 \rightarrow n = mg$$



ANS. FIG. P11.28

We must use the center of mass as the axis in

$$\sum \tau = I\alpha:$$
$$F_g\,(0) - n\,(77.5\text{ cm}) + f_s\,(88\text{ cm}) = 0$$

We combine the equations by substitution:

$$-mg(77.5\text{ cm}) + ma_x(88\text{ cm}) = 0$$
$$a_x = \frac{\left(9.80\text{ m/s}^2\right)77.5\text{ cm}}{88\text{ cm}} = \boxed{8.63\text{ m/s}^2}$$

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P11.29 We require $a_c = g = \dfrac{v^2}{r} = \omega^2 r$:

$$\omega = \sqrt{\frac{g}{r}} = \sqrt{\frac{\left(9.80\ \text{m/s}^2\right)}{100\ \text{m}}} = 0.313\ \text{rad/s}$$

$$I = Mr^2 = \left(5\times10^4\ \text{kg}\right)(100\ \text{m})^2 = 5\times10^8\ \text{kg}\cdot\text{m}^2$$

(a) $$L = I\omega = \left(5\times10^8\ \text{kg}\cdot\text{m}^2\right)(0.313\ \text{rad/s}) = \boxed{1.57\times10^8\ \text{kg}\cdot\text{m}^2/\text{s}}$$

(b) $$\Delta t = \frac{L_f - 0}{\sum\tau} = \frac{1.57\times10^8\ \text{kg}\cdot\text{m}^2/\text{s}}{2(125\ \text{N})(100\ \text{m})} = \boxed{6.26\times10^3\ \text{s}} = 1.74\ \text{h}$$

Section 11.4 Analysis Model: Isolated System (Angular Momentum)

P11.30 (a) From conservation of angular momentum for the isolated system of two disks:

$$\left(I_1 + I_2\right)\omega_f = I_1\omega_i \quad \text{or} \quad \omega_f = \boxed{\frac{I_1}{I_1 + I_2}\omega_i}$$

This is an example of a totally inelastic collision.

(b) $$K_f = \frac{1}{2}\left(I_1 + I_2\right)\omega_f^2 \qquad \text{and} \qquad K_i = \frac{1}{2}I_1\omega_i^2$$

so $$\frac{K_f}{K_i} = \frac{\frac{1}{2}\left(I_1 + I_2\right)}{\frac{1}{2}I_1\omega_i^2}\left(\frac{I_1}{I_1 + I_2}\omega_i\right)^2 = \boxed{\frac{I_1}{I_1 + I_2}}$$

P11.31 From conservation of angular momentum,

$$I_i\omega_i = I_f\omega_f\text{: } \left(250\ \text{kg}\cdot\text{m}^2\right)(10.0\ \text{rev/min}) =$$

$$\left[250\ \text{kg}\cdot\text{m}^2 + (25.0\ \text{kg})(2.00\ \text{m})^2\right]\omega_2$$

$$\omega_2 = \boxed{7.14\ \text{rev/min}}$$

P11.32 (a) Angular momentum is conserved in the puck-rod-putty system because there is no net external torque acting on the system.

$$I\omega_{\text{initial}} = I\omega_{\text{final}}\text{:}$$

$$mR^2\left(\frac{v_i}{R}\right) + m_pR^2(0) = \left(mR^2 + m_pR^2\right)\left(\frac{v_f}{R}\right)$$

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$$mRv_i = \left(m + m_p\right)Rv_f$$

Solving for the final velocity gives

$$v_f = \left(\frac{m}{m+m_p}\right)v_i = \left(\frac{2.40\text{ kg}}{2.40\text{ kg} + 1.30\text{ kg}}\right)(5.00\text{ m/s}) = 3.24\text{ m/s}$$

Then,

$$T = \frac{2\pi R}{v_f} = \frac{2\pi(1.50\text{ m})}{3.24\text{ m/s}} = \boxed{2.91\text{ s}}$$

(b) Yes, because there is no net external torque acting on the puck-rod-putty system.

(c) No, because the pivot pin is always pulling on the rod to change the direction of the momentum.

(d) No. Some mechanical energy is converted into internal energy. The collision is perfectly inelastic.

P11.33 (a) Mechanical energy is not constant; some chemical potential energy in the woman's body is transformed into mechanical energy.

(b) Momentum is not constant. The turntable bearing exerts an external northward force on the axle to prevent the axle from moving southward because of the northward motion of the woman.

(c) Angular momentum is constant because the system is isolated from torque about the axle.

(d) From conservation of angular momentum for the system of the woman and the turntable, we have $L_f = L_i = 0$,

so, $L_f = I_{\text{woman}}\omega_{\text{woman}} + I_{\text{table}}\omega_{\text{table}} = 0$

and $$\omega_{\text{table}} = \left(-\frac{I_{\text{woman}}}{I_{\text{table}}}\right)\omega_{\text{woman}} = \left(-\frac{m_{\text{woman}}r^2}{I_{\text{table}}}\right)\left(\frac{v_{\text{woman}}}{r}\right)$$

$$= -\frac{m_{\text{woman}}rv_{\text{woman}}}{I_{\text{table}}}$$

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$$\omega_{\text{table}} = -\frac{60.0\text{ kg}(2.00\text{ m})(1.50\text{ m/s})}{500\text{ kg}\cdot\text{m}^2} = -0.360\text{ rad/s}$$

or $\omega_{\text{table}} = \boxed{0.360\text{ rad/s (counterclockwise)}}$

(e) Chemical energy converted into mechanical energy is equal to

$$\Delta K = K_f - 0 = \frac{1}{2}m_{\text{woman}}v_{\text{woman}}^2 + \frac{1}{2}I\omega_{\text{table}}^2$$

$$\Delta K = \frac{1}{2}(60\text{ kg})(1.50\text{ m/s})^2 + \frac{1}{2}(500\text{ kg}\cdot\text{m}^2)(0.360\text{ rad/s})^2$$

$$= \boxed{99.9\text{ J}}$$

P11.34 (a) The total angular momentum of the system of the student, the stool, and the weights about the axis of rotation is given by

$$I_{\text{total}} = I_{\text{weights}} + I_{\text{student}} = 2(mr^2) + 3.00\text{ kg}\cdot\text{m}^2$$

Before: $r = 1.00\text{ m}$

Thus, $I_i = 2(3.00\text{ kg})(1.00\text{ m})^2 + 3.00\text{ kg}\cdot\text{m}^2 = 9.00\text{ kg}\cdot\text{m}^2$

After: $r = 0.300\text{ m}$

Thus, $I_f = 2(3.00\text{ kg})(0.300\text{ m})^2 + 3.00\text{ kg}\cdot\text{m}^2 = 3.54\text{ kg}\cdot\text{m}^2$

We now use conservation of angular momentum.

$$I_f\omega_f = I_i\omega_i$$

or $\omega_f = \left(\frac{I_i}{I_f}\right)\omega_i = \left(\frac{9.00}{3.54}\right)(0.750\text{ rad/s}) = \boxed{1.91\text{ rad/s}}$

(b) $K_i = \frac{1}{2}I_i\omega_i^2 = \frac{1}{2}(9.00\text{ kg}\cdot\text{m}^2)(0.750\text{ rad/s})^2 = \boxed{2.53\text{ J}}$

$K_f = \frac{1}{2}I_f\omega_f^2 = \frac{1}{2}(3.54\text{ kg}\cdot\text{m}^2)(1.91\text{ rad/s})^2 = \boxed{6.44\text{ J}}$

P11.35 (a) We solve by using conservation of angular momentum for the turntable-clay system, which is isolated from outside torques:

$I\omega_{\text{initial}} = I\omega_{\text{final}}$:

$$\frac{1}{2}mR^2\omega_i = \left(\frac{1}{2}mR^2 + m_c r^2\right)\omega_f$$

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Solving for the final angular velocity gives

$$\omega_f = \frac{\frac{1}{2}mR^2\omega_i}{\frac{1}{2}mR^2 + m_c r^2} = \frac{\frac{1}{2}(30.0\text{ kg})(1.90\text{ m})^2(4\pi\text{ rad/s})}{\frac{1}{2}(30.0\text{ kg})(1.90\text{ m})^2 + (2.25\text{ kg})(1.80\text{ m})^2}$$

$$= \boxed{11.1\text{ rad/s counterclockwise}}$$

(b) No. The initial energy is

$$K_i = \frac{1}{2}I\omega_i^2 = \frac{1}{2}\left(\frac{1}{2}mR^2\right)\omega_i^2$$

$$= \frac{1}{2}\left[\frac{1}{2}(30.0\text{ kg})(1.90\text{ m})^2\right](4\pi\text{ rad/s})^2$$

$$= 4\ 276\text{ J}$$

The final mechanical energy is

$$K_f = \frac{1}{2}I\omega_f^2 = \frac{1}{2}\left(\frac{1}{2}mR^2 + m_c r^2\right)\omega_f^2$$

$$= \frac{1}{2}\left[\frac{1}{2}(30.0\text{ kg})(1.90\text{ m})^2 + (2.25\text{ kg})(1.80\text{ m})^2\right] \times (11.1\text{ rad/s})^2$$

$$= 3\ 768\text{ J}$$

Thus 507 J of mechanical energy is transformed into internal energy. The "angular collision" is completely inelastic.

(c) No. The original horizontal momentum is zero. As soon as the clay has stopped skidding on the turntable, the final momentum is (2.25 kg)(1.80 m)(11.1 rad/s) = 44.9 kg · m/s north. This is the amount of impulse injected by the bearing. The bearing thereafter keeps changing the system momentum to change the direction of the motion of the clay. The turntable bearing promptly imparts an impulse of 44.9 kg · m/s north into the turntable-clay system, and thereafter keeps changing the system momentum.

P11.36 When they touch, the center of mass is distant from the center of the larger puck by

$$y_{CM} = \frac{0 + (80.0\text{ g})(4.00\text{ cm} + 6.00\text{ cm})}{120\text{ g} + 80.0\text{ g}} = 4.00\text{ cm}$$

(a) $L = r_1 m_1 v_1 + r_2 m_2 v_2 = 0 + (6.00\times10^{-2}\text{ m})(80.0\times10^{-3}\text{ kg})(1.50\text{ m/s})$

$$= \boxed{7.20\times10^{-3}\text{ kg}\cdot\text{m}^2/\text{s}}$$

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(b) The moment of inertia about the CM is

$$I = \left(\frac{1}{2}m_1 r_1^2 + m_1 d_1^2\right) + \left(\frac{1}{2}m_2 r_2^2 + m_2 d_2^2\right)$$

$$I = \frac{1}{2}(0.120\ \text{kg})(6.00\times10^{-2}\ \text{m})^2 + (0.120\ \text{kg})(4.00\times10^{-2})^2$$

$$+\frac{1}{2}(80.0\times10^{-3}\ \text{kg})(4.00\times10^{-2}\ \text{m})^2$$

$$+(80.0\times10^{-3}\ \text{kg})(6.00\times10^{-2}\ \text{m})^2$$

$$I = 7.60\times10^{-4}\ \text{kg}\cdot\text{m}^2$$

Angular momentum of the two-puck system is conserved:

$$L = I\omega$$

$$\omega = \frac{L}{I} = \frac{7.20\times10^{-3}\ \text{kg}\cdot\text{m}^2/\text{s}}{7.60\times10^{-4}\ \text{kg}\cdot\text{m}^2} = \boxed{9.47\ \text{rad/s}}$$

P11.37 (a) Taking the origin at the pivot point, note that $\vec{\mathbf{r}}$ is perpendicular to $\vec{\mathbf{v}}$, so $\sin\theta = 1$ and $L_f = L_i = mr\sin\theta = \boxed{mv\ell}$ vertically down.

(b) Taking v_f to be the speed of the bullet and the block together, we first apply conservation of angular momentum: $L_i = L_f$ becomes

$$\ell mv = \ell(m+M)v_f \quad \text{or} \quad v_f = \left(\frac{m}{m+M}\right)v$$

The total kinetic energies before and after the collision are, respectively,

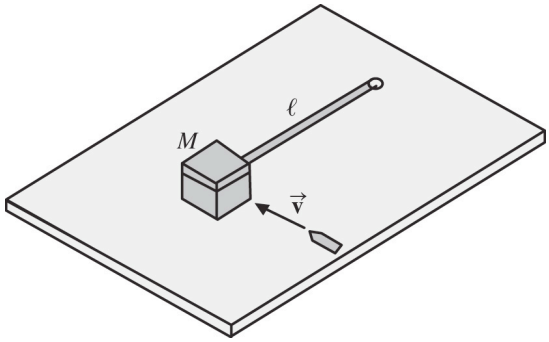
$$K_i = \frac{1}{2}mv^2$$

and $$K_f = \frac{1}{2}(m+M)v_f^2 = \frac{1}{2}(m+M)\left(\frac{m}{m+M}\right)^2 v^2 = \frac{1}{2}\left(\frac{m^2}{m+M}\right)v^2$$

So the fraction of the kinetic energy that is converted into internal energy will be

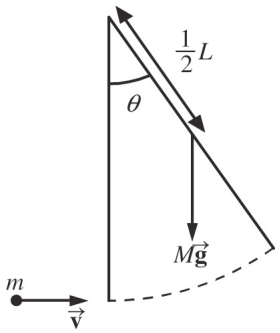
$$\text{Fraction} = \frac{-\Delta K}{K_i} = \frac{K_i - K_f}{K_i} = \frac{\frac{1}{2}mv^2 - \frac{1}{2}\left(\frac{m^2}{m+M}\right)v^2}{\frac{1}{2}mv^2} = \boxed{\frac{M}{m+M}}$$

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ANS. FIG. P11.37

P11.38 (a) Let ω be the angular speed of the signboard when it is vertical.



ANS. FIG. P11.38

$$\frac{1}{2}I\omega^2 = Mgh$$

$$\frac{1}{2}\left(\frac{1}{3}ML^2\right)\omega^2 = Mg\frac{1}{2}L(1-\cos\theta)$$

$$\omega = \sqrt{\frac{3g(1-\cos\theta)}{L}}$$

$$= \sqrt{\frac{3(9.80\ \text{m/s}^2)(1-\cos 25.0^\circ)}{0.500\ \text{m}}}$$

$$= \boxed{2.35\ \text{rad/s}}$$

(b) $I_i\omega_i - mvL = I_f\omega_f$ represents angular momentum conservation for the sign-snowball system. Substituting into the above equation,

$$\left(\frac{1}{3}ML^2 + mL^2\right)\omega_f = \frac{1}{3}ML^2\omega_i - mvL$$

Solving,

$$\omega_f = \frac{\frac{1}{3}ML\omega_i - mv}{\left(\frac{1}{3}M + m\right)L}$$

$$= \frac{\frac{1}{3}(2.40\ \text{kg})(0.500\ \text{m})(2.347\ \text{rad/s}) - (0.400\ \text{kg})(1.60\ \text{m/s})}{\left[\frac{1}{3}(2.40\ \text{kg}) + 0.400\ \text{kg}\right](0.500\ \text{m})}$$

$$= \boxed{0.498\ \text{rad/s}}$$

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(c) Let h_{CM} = distance of center of mass from the axis of rotation.

$$h_{CM} = \frac{(2.40\text{ kg})(0.250\text{ m}) + (0.400\text{ kg})(0.500\text{ m})}{2.40\text{ kg} + 0.400\text{ kg}} = 0.285\ 7\text{ m}$$

Applying conservation of mechanical energy,

$$(M+m)gh_{CM}(1-\cos\theta) = \frac{1}{2}\left(\frac{1}{3}ML^2 + mL^2\right)\omega^2$$

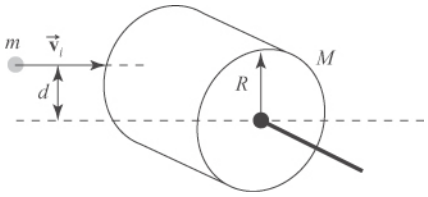
Solving for θ then gives

$$\theta = \cos^{-1}\left[1 - \frac{\left(\frac{1}{3}M + m\right)L^2\omega^2}{2(M+m)gh_{CM}}\right]$$

$$= \cos^{-1}\left\{1 - \frac{\left[\frac{1}{3}(2.40\text{ kg}) + 0.400\text{ kg}\right](0.500\text{ m})^2(0.498\text{ rad/s})^2}{2(2.40\text{ kg} + 0.400\text{ kg})(9.80\text{ m/s}^2)(0.285\ 7\text{ m})}\right\}$$

$$= \boxed{5.58°}$$

P11.39 (a) Consider the system to consist of the wad of clay and the cylinder. No external forces acting on this system have a torque about the center of the cylinder. Thus, angular momentum of the system is conserved about the axis of the cylinder.



ANS. FIG. P11.39

$$L_f = L_i: \qquad I\omega = mv_i d$$

or $$\left[\frac{1}{2}MR^2 + mR^2\right]\omega = mv_i d$$

Thus, $$\omega = \boxed{\frac{2mv_i d}{(M+2m)R^2}}$$

(b) No; some mechanical energy of the system (the kinetic energy of the clay) changes into internal energy.

(c) The linear momentum of the system is not constant. The axle exerts a backward force on the cylinder when the clay strikes.

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P11.40 The rotation rate of the station is such that at its rim the centripetal acceleration, a_c, is equal to the acceleration of gravity on the Earth's surface, g. Thus, the normal force from the rim's floor provides centripetal force on any person equal to that person's weight:

$$\sum F_r = ma_r: \quad n = \frac{mv^2}{r} \rightarrow mg = m\omega_i^2 r \rightarrow \omega_i^2 = \frac{g}{r}$$

The space station is isolated, so its angular momentum is conserved. When the people move to the center, the station's moment of inertia decreases, its angular speed increases, and the effective value of gravity increases.

From angular momentum conservation: $I_i\omega_i = I_f\omega_f \rightarrow \dfrac{\omega_f}{\omega_i} = \dfrac{I_i}{I_f}$, where

$$\begin{aligned} I_i &= I_{\text{station}} + I_{\text{people},\,i} \\ &= \left[5.00\times10^8\ \text{kg}\cdot\text{m}^2 + 150(65.0\ \text{kg})(100\ \text{m})^2\right] \\ &= 5.98\times10^8\ \text{kg}\cdot\text{m}^2 \end{aligned}$$

$$\begin{aligned} I_f &= I_{\text{station}} + I_{\text{people},\,f} \\ &= \left[5.00\times10^8\ \text{kg}\cdot\text{m}^2 + 50(65.0\ \text{kg})(100\ \text{m})^2\right] \\ &= 5.32\times10^8\ \text{kg}\cdot\text{m}^2 \end{aligned}$$

The centripetal acceleration is the effective value of gravity: $a_c \propto g$. Comparing values of acceleration before and during the union meeting, we have

$$\frac{g_f}{g_i} = \frac{a_{c,f}}{a_{c,i}} = \frac{\omega_f^2}{\omega_i^2} = \left(\frac{I_i}{I_f}\right)^2 = \left(\frac{5.98\times10^8}{5.32\times10^8}\right)^2 = 1.26 \rightarrow g_f = 1.26g_i$$

When the people move to the center, the angular speed of the station increases. This increases the effective gravity by 26%. Therefore, the ball will not take the same amount of time to drop.

P11.41 (a) Yes, the bullet has angular momentum about an axis through the hinges of the door before the collision.

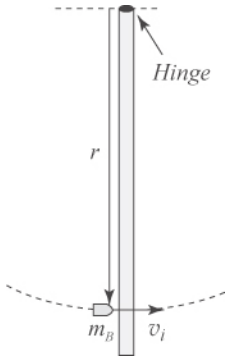
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(b) The bullet strikes the door

$$r = 1.00\text{ m} - 0.100\text{ m} = 0.900\text{ m}$$

from the hinge. Its initial angular momentum is therefore

$$\begin{aligned} L_i &= rp = m_B r v_i \\ &= (0.005\ 00\text{ kg})(0.900\text{ m}) \\ &\quad \times (1.00\times 10^3\text{ m/s}) \\ &= \boxed{4.50\text{ kg}\cdot\text{m}^2/\text{s}} \end{aligned}$$



ANS. FIG. P11.41

(c) No; in the perfectly inelastic collision kinetic energy is transformed to internal energy.

(d) Apply conservation of angular momentum, $L_i = L_f$:

$$m_B r v_i = I_f \omega_f = (I_{\text{door}} + I_{\text{bullet}})\omega_f$$

$$m_B r v_i = \left(\frac{1}{3}M_{\text{door}}L^2 + m_B r^2\right)\omega_f$$

where L = 1.00 m = the width of the door and r = 0.900 m [from part (b)]. Solving for the final angular velocity gives,

$$\begin{aligned} \omega &= \frac{m_B r v_i}{\frac{1}{3}M_{\text{door}}L^2 + m_B r^2} \\ &= \frac{(0.005\ 00\text{ kg})(0.900\text{ m})(1.00\times 10^3\text{ m/s})}{\frac{1}{3}(18.0\text{ kg})(1.00\text{ m})^2 + (0.005\ 00\text{ kg})(0.900\text{ m})^2} \\ &= \boxed{0.749\text{ rad/s}} \end{aligned}$$

(e) The kinetic energy of the door-bullet system immediately after impact is

$$\begin{aligned} KE_f &= \frac{1}{2}I_f\omega_f^2 \\ &= \frac{1}{2}\left[\frac{1}{3}(18.0\text{ kg})(1.00\text{ m})^2 + (0.005\ 00\text{ kg})(0.900\text{ m})^2\right] \\ &\quad \times (0.749\text{ rad/s})^2 \\ &= \boxed{1.68\text{ J}} \end{aligned}$$

The kinetic energy (of the bullet) just before impact was

$$KE_i = \frac{1}{2}m_B v_i^2 = \frac{1}{2}(0.005\ 00\text{ kg})(1.00\times 10^3\text{ m/s})^2 = 2.50\times 10^3\text{ J}$$

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The total energy of the system must be the same before and after the collision, assuming we ignore the energy leaving by mechanical waves (sound) and heat (from the newly-warmer door to the cooler air). The kinetic energies are as follows: $KE_i = 2.50 \times 10^3$ J and $KE_f = 1.68$ J.

Most of the initial kinetic energy is transformed to internal energy in the collision.

Section 11.5 The Motion of Gyroscopes and Tops

P11.42 Angular momentum of the system of the spacecraft and the gyroscope is conserved. The gyroscope and spacecraft turn in opposite directions.

$$0 = I_1\omega_1 + I_2\omega_2: \qquad -I_1\omega_1 = I_2\frac{\theta}{t}$$

$$\left(-20\ \text{kg}\cdot\text{m}^2\right)\left(-100\ \text{rad/s}\right) = \left(5\times10^5\ \text{kg}\cdot\text{m}^2\right)\left(\frac{30^\circ}{t}\right)\left(\frac{\pi\ \text{rad}}{180^\circ}\right)$$

$$t = \frac{2.62\times10^5\ \text{s}}{2000} = \boxed{131\ \text{s}}$$

P11.43 We begin by calculating the moment of inertia of the Earth, modeled as a sphere:

$$I = \frac{2}{5}MR^2 = \frac{2}{5}\left(5.98\times10^{24}\ \text{kg}\right)\left(6.37\times10^6\ \text{m}\right)^2$$
$$= 9.71\times10^{37}\ \text{kg}\cdot\text{m}^2$$

Earth's rotational angular momentum is then

$$L = I\omega = \left(9.71\times10^{37}\ \text{kg}\cdot\text{m}^2\right)\left(\frac{2\pi\ \text{rad}}{86\,400\ \text{s}}\right) = 7.06\times10^{33}\ \text{kg}\cdot\text{m}^2/\text{s}^2$$

from which we can calculate the torque that is causing the precession:

$$\tau = L\omega_p$$
$$= \left(7.06\times10^{33}\ \text{kg}\cdot\text{m}^2/\text{s}\right)\left(\frac{2\pi\ \text{rad}}{2.58\times10^4\ \text{yr}}\right)\left(\frac{1\ \text{yr}}{365.25\ \text{d}}\right)\left(\frac{1\ \text{d}}{86\,400\ \text{s}}\right)$$
$$= \boxed{5.45\times10^{22}\ \text{N}\cdot\text{m}}$$

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Additional Problems

P11.44 (a) Assuming the rope is massless, the tension is the same on both sides of the pulley:

$$\sum \tau = TR - TR = \boxed{0}$$

(b) $\sum \tau = \dfrac{dL}{dt}$, and since $\sum \tau = 0$, L = constant.

Since the total angular momentum of the system is initially zero, the total angular momentum remains zero, so the monkey and bananas move upward with the same speed at any instant.



ANS. FIG. P11.44

(c) The monkey will not reach the bananas.

The motions of the monkey and bananas are identical, so the bananas remain out of the monkey's reach—until they get tangled in the pulley. To state the evidence differently, the tension in the rope is the same on both sides. Newton's second law applied to the monkey and bananas give the same acceleration upward.

P11.45 Using conservation of angular momentum, we have

$$L_{\text{aphelion}} = L_{\text{perihelion}} \quad \text{or} \quad \left(mr_a^2\right)\omega_a = \left(mr_p^2\right)\omega_p$$

Thus, $\left(mr_a^2\right)\dfrac{v_a}{r_a} = \left(mr_p^2\right)\dfrac{v_p}{r_p}$, giving $r_a v_a = r_p v_p$ or

$$v_a = \frac{r_p}{r_a} v_p = \frac{0.590 \text{ AU}}{35.0 \text{ AU}}(54.0 \text{ km/s}) = \boxed{0.910 \text{ km/s}}$$

P11.46 (a) Momentum is conserved in the isolated system of the two boys:

$$\vec{\mathbf{p}}_i = \vec{\mathbf{p}}_f : m_1 v_1 \,\hat{\mathbf{i}} - m_2 v_2 \,\hat{\mathbf{i}} = \left(m_1 + m_2\right)\vec{\mathbf{v}}_f$$

$$\vec{\mathbf{p}}_i = (45.0 \text{ kg})(8.00 \text{ m/s})\,\hat{\mathbf{i}} - (31.0 \text{ kg})(11.0 \text{ m/s})\,\hat{\mathbf{i}}$$
$$= (76.0 \text{ kg})\,\vec{\mathbf{v}}_f$$

$$\vec{\mathbf{v}}_f = \boxed{0.250\,\hat{\mathbf{i}} \text{ m/s}}$$

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(b) The initial kinetic energy of the system is

$$K_i = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$$
$$= \frac{1}{2}(45.0\text{ kg})(+8.00\text{ m/s})^2 + \frac{1}{2}(31.0\text{ kg})(11.0\text{ m/s})^2$$
$$= 3\ 315.5\text{ J}$$

and the kinetic energy after the collision is

$$K_f = \frac{1}{2}(m_1 + m_2)v_f^2 = \frac{1}{2}(76.0\text{ kg})(0.250\text{ m/s})^2 = 2.375\text{ J}$$

Thus the fraction remaining is

$$\frac{K_f}{K_i} = \frac{2.375\text{ J}}{3\ 315.5\text{ J}} = \boxed{0.000\ 716} = 0.071\ 6\%$$

(c) The calculation in part (a) still applies: $\vec{\mathbf{v}}_f = \boxed{0.250\ \hat{\mathbf{i}}\text{ m/s}}$

(d) Taking Jacob (m_1 = 45.0 kg) at the origin of a coordinate system, with Ethan (m_2 = 31.0 kg) on the y axis at $y = L$ = 1.20 m, the position of the CM of the boys is

$$y_{\text{CM}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{m_1(0) + m_2 L}{m_1 + m_2} = \frac{m_2 L}{m_1 + m_2}$$
$$y_{\text{CM}} = \frac{(31.0\text{ kg})(1.20\text{ m})}{45.0\text{ kg} + 31.0\text{ kg}} = 0.489\text{ m}$$

Jacob is y_{CM} = 0.489 m from the CM and Ethan is $(L - y_{\text{CM}})$ = $L - m_2 L/(m_1 + m_2) = m_1 L/(m_1 + m_2) = 0.711\text{ m}$ from the CM. Their angular momentum about the CM is $L = I\omega$:

$$m_1 v_1 L + m_2 v_2 (L - y_{\text{CM}}) = \left[m_1 L^2 + m_2 (L - y_{\text{CM}})^2 \right]\omega$$
$$\rightarrow \omega = \frac{m_1 v_1 L + m_2 v_2 (L - y_{\text{CM}})}{m_2 L^2 + m_2 (L - y_{\text{CM}})^2}$$

$$\omega = \frac{(45.0\text{ kg})(8.00\text{ m/s})(0.489\text{ m}) + (31.0\text{ kg})(11.0\text{ m/s})(0.711\text{ m})}{(45.0\text{ kg})(0.489\text{ m})^2 + (31.0\text{ kg})(0.711\text{ m})^2}$$

$$\omega = \frac{418\text{ kg}\cdot\text{m}^2/\text{s}}{26.4\text{ kg}\cdot\text{m}^2} = \boxed{15.8\text{ rad/s}}$$

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(e) Their kinetic energy after they link arms is

$$K_f = \frac{1}{2}(m_1 + m_2)v_{\text{CM}}^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}(76.0 \text{ kg})(0.250 \text{ m/s})^2 + \frac{1}{2}(26.4 \text{ kg}\cdot\text{m}^2/\text{s})(15.8 \text{ rad/s})^2$$

$$K_f = 3\ 315.5 \text{ J}$$

[Note: the result of the calculation of kinetic energy is exactly 3 315.5 J if no round-off is made in the calculation. It can be shown algebraically that the expression for the final kinetic energy is equivalent to the expression for the initial kinetic energy—the student is invited to show this.] Thus the fraction remaining is $K_f/K_i = 3\ 315.5 \text{ J}/3\ 315.5 \text{ J} = \boxed{1.00} = 100\%$.

(f) In part (b), the boys must necessarily deform as they slam into each other. During this deformation process, mechanical energy is transformed into internal energy. In part (e), there is no deformation involved. The boys simply link hands and some of their translational kinetic energy transforms to rotational kinetic energy, but none is transformed to internal energy.

P11.47 First, we define the following symbols:

I_P = moment of inertia due to mass of people on the equator

I_E = moment of inertia of the Earth alone (without people)

ω = angular velocity of the Earth (due to rotation on its axis)

$T = \dfrac{2\pi}{\omega}$ = rotational period of the Earth (length of the day)

R = radius of the Earth

The initial angular momentum of the system (before people start running) is

$$L_i = I_P\omega_i + I_E\omega_i = (I_P + I_E)\omega_i$$

When the Earth has angular speed ω, the tangential speed of a point on the equator is $v_t = R\omega$. Thus, when the people run eastward along the equator at speed v relative to the surface of the Earth, their tangential speed is $v_p = v_t + v = R\omega + v$ and their angular speed is

$$\omega_P = \frac{v_p}{R} = \omega + \frac{v}{R}$$

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The angular momentum of the system after the people begin to run is

$$L_f = I_P\omega_p + I_E\omega = I_P\left(\omega + \frac{v}{R}\right) + I_E\omega = \left(I_P + I_E\right)\omega + \frac{I_P v}{R}$$

Since no external torques have acted on the system, angular momentum is conserved $\left(L_f = L_i\right)$, giving

$$\left(I_P + I_E\right)\omega + \frac{I_P v}{R} = \left(I_P + I_E\right)\omega_i$$

Thus, the final angular velocity of the Earth is

$$\omega = \omega_i - \frac{I_P v}{\left(I_P + I_E\right)R} = \omega_i(1-x), \text{ where } x \equiv \frac{I_P v}{\left(I_P + I_E\right)R\omega_i}$$

The new length of the day is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\omega_i(1-x)} = \frac{T_i}{1-x} \approx T_i(1+x)$$

so the increase in the length of the day is

$$\Delta T = T - T_i \approx T_i x = T_i\left[\frac{I_P v}{\left(I_P + I_E\right)R\omega_i}\right]$$

Since $\omega_i = \dfrac{2\pi}{T_i}$, this may be written as

$$\Delta T \approx \frac{T_i^2 I_P v}{2\pi\left(I_P + I_E\right)R}$$

To obtain a numeric answer, we compute

$$\begin{aligned} I_P &= m_p R^2 = \left[\left(7\times10^9\right)\left(55.0\text{ kg}\right)\right]\left(6.37\times10^6\text{ m}\right)^2 \\ &= 1.56\times10^{25}\text{ kg}\cdot\text{m}^2 \end{aligned}$$

and

$$\begin{aligned} I_E &= \frac{2}{5}m_E R^2 = \frac{2}{5}\left(5.98\times10^{24}\text{ kg}\right)\left(6.37\times10^6\text{ m}\right)^2 \\ &= 9.71\times10^{37}\text{ kg}\cdot\text{m}^2 \end{aligned}$$

Thus,

$$\begin{aligned} \Delta T &\approx \frac{\left(8.64\times10^4\text{ s}\right)^2\left(1.56\times10^{25}\text{ kg}\cdot\text{m}^2\right)(2.5\text{ m/s})}{2\pi\left[\left(1.56\times10^{25} + 9.71\times10^{37}\right)\text{ kg}\cdot\text{m}^2\right]\left(6.37\times10^6\text{ m}\right)} \\ &= \boxed{7.50\times10^{-11}\text{ s}} \end{aligned}$$

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P11.48 (a) $\left(K+U_g\right)_A=\left(K+U_g\right)_B$

$$0+mgy_A=\frac{1}{2}mv_B^2+0$$

$$v_B=\sqrt{2gy}_A=\sqrt{2\left(9.80\text{ m/s}^2\right)(6.30\text{ m})}=\boxed{11.1\text{ m/s}}$$

(b) $L=mvr=(76.0\text{ kg})\,(11.1\text{ m/s})\,(6.30\text{ m})=\boxed{5.32\times10^3\text{ kg}\cdot\text{m}^2/\text{s}}$ toward you along the axis of the channel.

(c) The wheels on his skateboard prevent any tangential force from acting on him. Then no torque about the axis of the channel acts on him and his angular momentum is constant. His legs convert chemical into mechanical energy. They do work to increase his kinetic energy. The normal force acts in the upward direction, perpendicular to the direction of motion of the skateboarder.

(d) $L=mvr$: $\quad v=\dfrac{5.32\times10^3\text{ kg}\cdot\text{m}^2/\text{s}}{(76.0\text{ kg})\,(5.85\text{ m})}=\boxed{12.0\text{ m/s}}$

(e) $\left(K+U_g\right)_B+U_{\text{chemical},B}=\left(K+U_g\right)_C$

$$\frac{1}{2}(76.0\text{ kg})(11.1\text{ m/s})^2+0+U_{\text{chem}}$$

$$=\frac{1}{2}(76.0\text{ kg})(12.0\text{ m/s})^2+(76.0\text{ kg})\,(9.80\text{ m/s}^2)\,(0.450\text{ m})$$

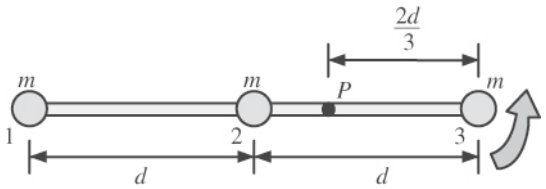
$$U_{\text{chem}}=5.44\text{ kJ}-4.69\text{ kJ}+335\text{ J}=\boxed{1.08\text{ kJ}}$$

P11.49 (a) The moment of inertia is given by

$$I=\sum m_i r_i^2$$

$$=m\left(\frac{4d}{3}\right)^2+m\left(\frac{d}{3}\right)^2+m\left(\frac{2d}{3}\right)^2$$

$$=\boxed{7m\frac{d^2}{3}}$$



ANS. FIG. P11.49

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(b) Think of the whole weight, $3mg$, acting at the center of gravity.

$$\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}} = \left(\frac{d}{3}\right)\left(-\hat{\mathbf{i}}\right) \times 3mg\left(-\hat{\mathbf{j}}\right) = \boxed{(mgd)\hat{\mathbf{k}}}$$

(c) We find the angular acceleration from

$$\alpha = \frac{\tau}{I} = \frac{3mgd}{7md^2} = \boxed{\frac{3g}{7d} \text{ counterclockwise}}$$

(d) The linear acceleration of particle 3, a distance of $2d/3$ from the pivot, is

$$a = \alpha r_3 = \left(\frac{3g}{7d}\right)\left(\frac{2d}{3}\right) = \boxed{\frac{2g}{7} \text{ upward}}$$

(e) Because the axle is fixed, no external work is performed on the system of the Earth and the three particles, so total mechanical energy is conserved. Rotational kinetic energy will be maximum when the rod has swung to a vertical orientation with the center of gravity directly under the axle. Take gravitational potential energy to be zero when the rod is in its vertical orientation. In the initial horizontal orientation, the center of gravity of the system will be $d/3$ higher:

$$E = (K+U)_{i\,=\,\text{horizontal}} = (K+U)_{f\,=\,\text{vertical}}$$

$$0 + (3m)g\left(\frac{d}{3}\right) = K_f + 0 \rightarrow K_f = \boxed{mgd}$$

(f) In the vertical orientation, the rod has the greatest rotational kinetic energy:

$$K_f = \frac{1}{2} I\omega_f^2$$

$$mgd = \frac{1}{2}\left(7m\frac{d^2}{3}\right)\omega_f^2 \rightarrow \omega_f = \boxed{\sqrt{\frac{6g}{7d}}}$$

(g) The maximum angular momentum of the system is

$$L_f = I\omega_f = \frac{7md^2}{3}\sqrt{\frac{6g}{7d}} = \boxed{\left(\frac{14g}{3}\right)^{1/2} md^{3/2}}$$

(h) The maximum speed of particle 2 is

$$v_f = \omega_f r_2 = \sqrt{\frac{6g}{7d}}\,\frac{d}{3} = \boxed{\sqrt{\frac{2gd}{21}}}$$

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P11.50 (a) The equation simplifies to

$$(1.75\ \text{kg}\cdot\text{m}^2/\text{s} - 0.181\ \text{kg}\cdot\text{m}^2/\text{s})\,\hat{\mathbf{j}} = (0.745\ \text{kg}\cdot\text{m}^2)\vec{\omega}$$

which gives

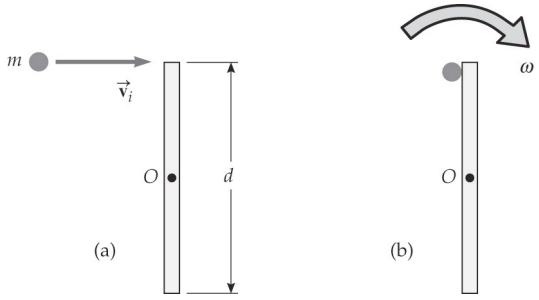
$$\vec{\omega} = \boxed{2.11\hat{\mathbf{j}}\ \text{rad/s}}$$

(b) We take the x axis east, the y axis up, and the z axis south.

The child has moment of inertia 0.730 kg·m^2 about the axis of the stool and is originally turning counterclockwise at 2.40 rad/s. At a point 0.350 m to the east of the axis, he catches a 0.120-kg ball moving toward the south at 4.30 m/s. He continues to hold the ball in his outstretched arm. Find his final angular velocity.

(c) Yes, with the left-hand side representing the final situation and the right-hand side representing the original situation, the equation describes the throwing process.

P11.51 (a) The appropriate model is to treat the projectile and the rod as an isolated system, experiencing no net external torque, or force.



ANS. FIG. P11.51

(b) $L_{total} = L_{particle} + L_{rod}$

$$= \frac{mv_i d}{2} + 0 = \boxed{\frac{mv_i d}{2}}$$

(c) $I_{total} = I_{particle} + I_{rod} = \frac{1}{12}Md^2 + m\left(\frac{d}{2}\right)^2$

$$I_{total} = \boxed{\frac{d^2(M+3m)}{12}}$$

(d) After the collision, we could express the angular momentum as,

$$L_{total} = I_{total}\omega = \boxed{\left(\frac{d^2(M+3m)}{12}\right)\omega}$$

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(e) Recognizing that angular momentum is conserved,

$$L_f = L_i$$

$$\left(\frac{d^2(M+3m)}{12}\right)\omega = \frac{mv_i d}{2}$$

$$\omega = \boxed{\frac{6mv_i}{d(M+3m)}}$$

(f) $$K = \boxed{\frac{1}{2}mv_i^2}$$

(g) $$K_{total} = \frac{1}{2}I_{total}\omega^2 = \frac{1}{2}\left(\frac{d^2(M+3m)}{12}\right)\left(\frac{6mv_i}{d(M+3m)}\right)^2$$

$$K_{total} = \boxed{\frac{3m^2v_i^2}{2(M+3m)}}$$

(h) The change in mechanical energy is,

$$|\Delta K| = \frac{1}{2}mv_i^2 - \frac{3m^2v_i^2}{2(M+3m)} = \frac{mMv_i^2}{2(M+3m)}$$

Then, the fractional change in the mechanical energy is

$$\frac{\frac{mMv_i^2}{2(M+3m)}}{\frac{1}{2}mv_i^2} = \boxed{\frac{M}{M+3m}}$$

P11.52 (a) The puck's linear momentum is always changing. Its mechanical energy changes as work is done on it. But its angular momentum stays constant because although an external force (the tension of the rope) acts on the puck, no external torques act.

Therefore, L = constant, and at any time,

$$mvr = mv_i r_i$$

giving us

$$v = \frac{v_i r_i}{r} = \frac{(1.50\ \text{m/s})(0.300\ \text{m})}{0.100\ \text{m}} = \boxed{4.50\ \text{m/s}}$$

(b) From Newton's second law, the tension is always

$$T = \frac{mv^2}{r} = \frac{(0.050\ 0\ \text{kg})(4.50\ \text{m/s})^2}{0.100\ \text{m}} = \boxed{10.1\ \text{N}}$$

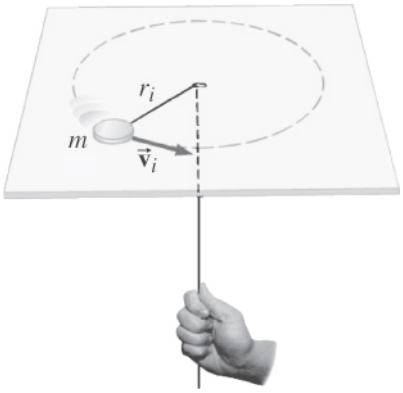
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(c) The work-kinetic energy theorem identifies the work as

$$W = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mv_i^2$$

$$= \frac{1}{2}(0.050\ 0\ \text{kg})\left[(4.50\ \text{m/s})^2 - (1.50\ \text{m/s})^2\right]$$

$$= \boxed{0.450\ \text{J}}$$



ANS. FIG. P11.52

P11.53 See ANS. FIG. P11.52 above.

(a) The puck is rotationally isolated because friction is zero and the torque on the puck from the tension in the string is zero:

$$\tau = \left|\vec{\mathbf{r}} \times \vec{\mathbf{F}}\right| = \left|\vec{\mathbf{r}}\right|\left|\vec{\mathbf{F}}\right| \sin 180^\circ = 0$$

therefore, the angular momentum of the puck is conserved as the radius is decreased:

$$L_f = L_i$$

$$mrv = mr_i v_i$$

$$\rightarrow \quad v = \boxed{\frac{r_i v_i}{r}}$$

(b) The net force on the puck is tension:

$$T = \frac{mv^2}{r} = \boxed{\frac{m(r_i v_i)^2}{r^3}}$$

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(c) Work is done by the tension force in the *negative-r*, inward direction as the radius decreases [$d\ell = -dr$]:

METHOD 1:

$$W = \int F \cdot d\ell = -\int T dr' = -\int_{r_i}^{r} \frac{m(r_i v_i)^2}{(r')^3} dr' = \left. \frac{m(r_i v_i)^2}{2(r')^2} \right|_{r_i}^{r}$$

$$= \frac{m(r_i v_i)^2}{2}\left(\frac{1}{r^2} - \frac{1}{r_i^2}\right) = \boxed{\frac{1}{2}mv_i^2\left(\frac{r_i^2}{r^2} - 1\right)}$$

METHOD 2:

$$W = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mv_i^2 = \boxed{\frac{1}{2}mv_i^2\left(\frac{r_i^2}{r^2} - 1\right)}$$

P11.54 The description of the problem allows us to assume the asteroid-Earth system is isolated, so angular momentum is conserved ($L_i = L_f$). Let the period of rotation of Earth be *T* before the collision and $T + \Delta T$ after the collision. We have

$$I_E \omega_i = (I_E + I_A)\omega_f$$

$$\frac{2\pi}{T} I_E = \frac{2\pi}{T + \Delta T}(I_E + I_A)$$

$$\frac{T + \Delta T}{T} = \frac{I_E + I_A}{I_E}$$

which gives

$$\frac{\Delta T}{T} = \frac{I_A}{I_E} \quad \rightarrow \quad I_A = I_E \frac{\Delta T}{T}$$

Treating Earth as a solid sphere of mass *M* and radius *R*, its moment of inertia is $\frac{2}{5}MR^2$. The moment of inertia of the asteroid at the equator is mR^2. We have then

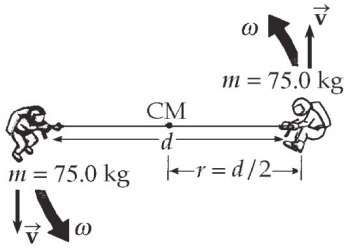
$$I_A = I_E \frac{\Delta T}{T} \quad \rightarrow \quad mR^2 = \left(\frac{2}{5}MR^2\right)\left(\frac{\Delta T}{T}\right) \quad \rightarrow \quad m = \frac{2}{5}M\left(\frac{\Delta T}{T}\right)$$

$$m = \frac{2}{5}(5.98 \times 10^{24}\ \text{kg})\left(\frac{0.500\ \text{s}}{24(3\ 600\ \text{s})}\right) = 1.38 \times 10^{19}\ \text{kg}$$

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Life would not go on as normal. An asteroid that would cause a 0.5-s change in the rotation period of the Earth has a mass of 1.38×10^{19} kg and is an order of magnitude larger in diameter than the one that caused the extinction of the dinosaurs.

P11.55 Both astronauts will speed up equally as angular momentum for the two-astronaut-rope system is conserved in the absence of external torques. We use this principle to find the new angular speed with the shorter tether. Standard equations will tell us the original amount of angular momentum and the original and final amounts of kinetic energy. Then the kinetic energy difference is the work.



ANS. FIG. P11.55

(a) The angular momentum magnitude is $\left|\vec{\mathbf{L}}\right| = m\left|\vec{\mathbf{r}}\times\vec{\mathbf{v}}\right|$. In this case, $\vec{\mathbf{r}}$ and $\vec{\mathbf{v}}$ are perpendicular, so the magnitude of L about the center of mass is

$$L = \sum mrv = 2(75.0\text{ kg})(5.00\text{ m})(5.00\text{ m/s})$$
$$= \boxed{3.75\times10^{3}\text{ kg}\cdot\text{m}^2/\text{s}}$$

(b) The original kinetic energy is

$$K = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 = 2\left(\frac{1}{2}\right)(75.0\text{ kg})(5.00\text{ m/s})^2$$
$$= \boxed{1.88\times10^{3}\text{ J}}$$

(c) With a lever arm of zero, the rope tension generates no torque about the center of mass. Thus, the angular momentum for the two-astronaut-rope system is unchanged:

$$L = \boxed{3.75\times10^{3}\text{ kg}\cdot\text{m}^2/\text{s}}$$

(d) Again, $L = 2mrv$, so

$$v = \frac{L}{2mr} = \frac{3.75\times10^{3}\text{ kg}\cdot\text{m}^2/\text{s}}{2(75.0\text{ kg})(2.50\text{ m})} = \boxed{10.0\text{ m/s}}$$

(e) The final kinetic energy is

$$K = 2\left(\frac{1}{2}mv^2\right) = 2\left(\frac{1}{2}\right)(75.0\text{ kg})(10.0\text{ m/s})^2 = \boxed{7.50\times10^{3}\text{ J}}$$

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(f) The energy converted by the astronaut is the work he does:

$$W_{nc} = K_f - K_i = 7.50\times10^3\ \text{J} - 1.88\times10^3\ \text{J}$$
$$= \boxed{5.62\times10^3\ \text{J}}$$

P11.56 Please refer to ANS. FIG. P11.55 and the discussion in P11.55 above.

(a) $L_i = 2\left[Mv\left(\frac{d}{2}\right)\right] = \boxed{Mvd}$

(b) $K = 2\left(\frac{1}{2}Mv^2\right) = \boxed{Mv^2}$

(c) $L_f = L_i = \boxed{Mvd}$

(d) $v_f = \frac{L_f}{2Mr_f} = \frac{Mvd}{2M\left(\frac{d}{4}\right)} = \boxed{2v}$

(e) $K_f = 2\left(\frac{1}{2}Mv_f^2\right) = M(2v)^2 = \boxed{4Mv^2}$

(f) If the work performed by the astronaut is made possible entirely by the conversion of chemical energy to mechanical energy, then the necessary chemical potential energy is:

$$W = K_f - K_i = \boxed{3Mv^2}$$

P11.57 (a) At the moment of release, two stones are moving with speed v_0. The total momentum has magnitude $\boxed{2mv_0}$. It keeps this same horizontal component of momentum as it flies away.

(b) The center of mass speed relative to the hunter is $v_{CM} = p/M = 2mv_0/3m = \boxed{2\,v_0/3}$ before the hunter lets go and, as far as horizontal motion is concerned, afterward.

(c) When the bola is first released, the stones are horizontally in line with two at distance ℓ on one side of the center knot and one at distance ℓ on the other side. The center of mass (CM) is then $x_{CM} = (2m\ell - m\ell)/3m = \ell/3$ from the center knot closer to the two stones: the one stone just being released is at distance $r_1 = 4\ell/3$ from the CM, the other two stones are at distance $r_2 = 2\ell/3$ from the CM.

The two stones, moving at v_0, have a relative speed $v_2 = v_0 - 2v_0/3 = v_0/3$ with respect to the CM, and the one stone has relative

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speed $v_1 = 2v_0/3 - 0 = 2v_0/3$ with respect to the CM. The one stone has angular speed

$$\omega_1 = \frac{v_1}{r_1} = \frac{2v_0/3}{4\ell/3} = \frac{v_0}{2\ell}$$

The other two stones have angular speed

$$\omega_2 = \frac{v_2}{r_2} = \frac{v_0/3}{2\ell/3} = \frac{v_0}{2\ell}$$

which is necessarily the same as that of stone 1: $\omega_1 = \omega_2 = \omega$. The total angular momentum around the center of mass is

$$\begin{aligned}\sum mvr &= mv_1r_1 + 2mv_2r_2 \\ &= m(2v_0/3)(4\ell/3) + 2m(v_0/3)(2\ell/3) \\ &= \boxed{4m\ell v_0/3}\end{aligned}$$

The angular momentum remains constant with this value as the bola flies away.

(d) As computed in part (c), the angular speed ω at the moment of release is $v_0/2\ell$. As it moves through the air, the bola keeps constant angular momentum, but its moment of inertia changes to $3m\ell^2$. Then the new angular speed is given by

$$L = I\omega \quad \rightarrow \quad 4m\ell v_0/3 = 3m\ell^2\omega \quad \rightarrow \quad \omega = \boxed{4v_0/9\ell}$$

(e) At the moment of release,

$$K = \frac{1}{2}m(0)^2 + \frac{1}{2}(2m)v_0^2 = \boxed{mv_0^2}$$

(f) As it flies off in its horizontal motion it has kinetic energy

$$\begin{aligned}K &= \frac{1}{2}(3m)\left(v_{\text{CM}}\right)^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}(3m)\left(\frac{2v_0}{3}\right)^2 + \frac{1}{2}\left(3m\ell^2\right)\left(\frac{4v_0}{9\ell}\right)^2 \\ &= \boxed{\frac{26}{27}mv_0^2}\end{aligned}$$

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(g) No horizontal forces act on the bola from outside after release, so the horizontal momentum stays constant. Its center of mass moves steadily with the horizontal velocity it had at release. No torques about its axis of rotation act on the bola, so its spin angular momentum stays constant. Internal forces cannot affect momentum conservation and angular momentum conservation, but they can affect mechanical energy. The cords pull on the stones as the stones rearrange themselves, so the cords must stretch slightly, so that energy of $mv_0^2/27$ changes from mechanical energy into internal energy as the bola takes its stable configuration. In a real situation, air resistance would have an influence on the motion of the stones.

P11.58 (a) Let M = mass of rod and m = mass of each bead. From $I_i\omega_i = I_f\omega_f$ between the moment of release and the moment the beads slide off, we have

$$\left[\frac{1}{12}M\ell^2 + 2mr_1^2\right]\omega_i = \left[\frac{1}{12}M\ell^2 + 2mr_2^2\right]\omega_f$$

When $M = 0.300$ kg, $\ell = 0.500$ m , $r_1 = 0.100$ m, $r_2 = 0.250$ m, and $\omega_i = 36.0$ rad/s, we find

$$[0.006\ 25 + 0.020\ 0m](36.0\ \text{rad/s}) = [0.006\ 25 + 0.125m]\omega_f$$

$$\boxed{\omega_f = \frac{36.0(1+3.20m)}{1+20.0m}\ \text{rad/s}}$$

(b) The denominator of this fraction always exceeds the numerator, so

ω_f decreases smoothly from a maximum value of 36.0 rad/s for $m = 0$ toward a minimum value of $(36 \times 3.2/20) = 5.76$ rad/s as $m \to \infty$.

P11.59 The moment of inertia of the rest of the Earth is

$$I = \frac{2}{5}MR^2 = \frac{2}{5}\left(5.98\times10^{24}\ \text{kg}\right)\left(6.37\times10^{6}\ \text{m}\right)^2$$
$$= 9.71\times10^{37}\ \text{kg}\cdot\text{m}^2$$

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For the original ice disks,

$$I = \frac{1}{2}Mr^2 = \frac{1}{2}\left(2.30\times10^{19}\ \text{kg}\right)\left(6\times10^{5}\ \text{m}\right)^2$$
$$= 4.14\times10^{30}\ \text{kg}\cdot\text{m}^2$$

For the final thin shell of water,

$$I = \frac{2}{3}Mr^2 = \frac{2}{3}\left(2.30\times10^{19}\ \text{kg}\right)\left(6.37\times10^{6}\ \text{m}\right)^2$$
$$= 6.22\times10^{32}\ \text{kg}\cdot\text{m}^2$$

Conservation of angular momentum for the spinning planet is expressed by $I_i\omega_i = I_f\omega_f$:

$$\left(4.14\times10^{30} + 9.71\times10^{37}\right)\frac{2\pi}{86\,400\ \text{s}}$$
$$= \left(6.22\times10^{32} + 9.71\times10^{37}\right)\frac{2\pi}{\left(86\,400\ \text{s} + \delta T\right)}$$

$$\left(1+\frac{\delta T}{86\,400\ \text{s}}\right)\left(1+\frac{4.14\times10^{30}}{9.71\times10^{37}}\right) = 1+\frac{6.22\times10^{32}}{9.71\times10^{37}}$$

$$\frac{\delta T}{86\,400\ \text{s}} = \frac{6.22\times10^{32}}{9.71\times10^{37}} - \frac{4.14\times10^{30}}{9.71\times10^{37}} \rightarrow \delta T = 0.550\ \text{s}$$

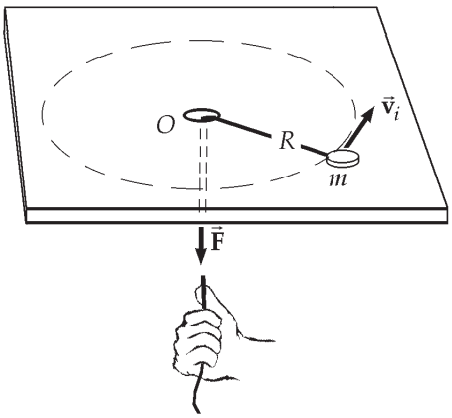
An increase of $6.368\times10^{-4}\,\%$ or 0.550 s.

P11.60 To evaluate the change in kinetic energy of the puck, we first calculate the initial and final moments of inertia of the puck:

$$I_i = mr_i^2$$
$$= (0.120\ \text{kg})(0.400\ \text{m})^2$$
$$= 1.92\times10^{-2}\ \text{kg}\cdot\text{m}^2$$

and

$$I_f = mr_f^2$$
$$= (0.120\ \text{kg})(0.250\ \text{m})^2$$
$$= 7.50\times10^{-3}\ \text{kg}\cdot\text{m}^2$$



ANS. FIG. P11.60

The initial angular velocity of the puck is given by

$$\omega_i = \frac{v_i}{r_i} = \frac{0.800\ \text{m/s}}{0.400\ \text{m}} = 2.00\ \text{rad/s}$$

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Now, use conservation of angular momentum for the system of the puck,

$$\omega_f = \omega_i\left(\frac{I_i}{I_f}\right) = (2.00\ \text{rad/s})\left(\frac{1.92\times10^{-2}\ \text{kg}\cdot\text{m}^2}{7.5\times10^{-3}\ \text{kg}\cdot\text{m}^2}\right) = 5.12\ \text{rad/s}$$

Now,

$$\begin{aligned}\text{work done} = \Delta K &= \frac{1}{2}I_f\omega_f^2 - \frac{1}{2}I_i\omega_i^2 \\ &= \frac{1}{2}(7.50\times10^{-3}\ \text{kg}\cdot\text{m}^2)(5.12\ \text{rad/s})^2 \\ &\qquad -\frac{1}{2}(1.92\times10^{-2}\ \text{kg}\cdot\text{m}^2)(2.00\ \text{rad/s})^2 \\ &= \boxed{5.99\times10^{-2}\ \text{J}}\end{aligned}$$

Challenge Problems

P11.61 (a) From the particle under a net force model:

$$F = \frac{\Delta p}{\Delta t} \quad\rightarrow\quad f_k = \frac{m\left(v_f - 0\right)}{\Delta t} = \frac{mv_f}{\Delta t} \qquad \textbf{[1]}$$

and from the rigid object under a net torque model:

$$\tau = \frac{\Delta L}{\Delta t} \quad\rightarrow\quad -f_k R = \frac{I\left(\omega_f - \omega_i\right)}{\Delta t} \qquad \textbf{[2]}$$

Divide [2] by [1]:

$$-R = \frac{I\left(\omega_f - \omega_i\right)}{mv_f}$$

Let $v_f = R\omega_f$ for pure rolling:

$$-R = \frac{I\left(\omega_f - \omega_i\right)}{m\left(R\omega_f\right)}$$

Solve for ω_f:

$$\omega_f = \frac{I\omega_i}{I + mR^2} = \frac{\frac{1}{2}mR^2\omega_i}{\frac{1}{2}mR^2 + mR^2} = \frac{\frac{1}{2}mR^2\omega_i}{\frac{3}{2}mR^2} = \boxed{\frac{1}{3}\omega_i}$$

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(b) The fractional change in kinetic energy is

$$\frac{\Delta E}{E} = \frac{\frac{1}{2}I\omega_f^2 + \frac{1}{2}Mv_{\text{CM}}^2 - \frac{1}{2}I\omega_i^2}{\frac{1}{2}I\omega_i^2}$$

$$= \frac{\frac{1}{2}\left(\frac{1}{2}MR^2\right)\left(\omega_i/3\right)^2 + \frac{1}{2}M\left(R\omega_i/3\right)^2 - \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega_i^2}{\frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega_i^2}$$

$$= \boxed{-\frac{2}{3}}$$

(c) $$\Delta t = \frac{\Delta p}{f} = \frac{Mv_f}{\mu Mg} = \frac{MR\omega_f}{\mu Mg} = \boxed{\frac{R\omega_f}{3\mu g}}$$

(d) From the particle under constant acceleration model:

$$\Delta x = v_{\text{avg}}\Delta t = \frac{0 + v_f}{2}\Delta t = \frac{1}{2}v_f\Delta t = \frac{1}{2}\left(R\omega_f\right)\left[\frac{1}{3}\left(\frac{R\omega_i}{\mu g}\right)\right]$$

$$= \frac{1}{2}\left[R\left(\frac{1}{3}\omega_i\right)\right]\left[\frac{1}{3}\left(\frac{R\omega_i}{\mu g}\right)\right] = \boxed{\frac{R^2\omega_i^2}{18\mu g}}$$

P11.62 (a) After impact, the disk adheres to the stick, so they will rotate about their common center of mass; therefore, we must consider the angular momentum of the system about its CM. First we find the velocity of the CM by writing the equations for momentum conservation:

$$m_d v_{di} + 0 = \left(m_d + m_s\right)v_{\text{CM}}$$

$$v_{\text{CM}} = \frac{m_d}{m_d + m_s}v_{di} = \left(\frac{2.0\text{ kg}}{2.0\text{ kg} + 1.0\text{ kg}}\right)(3.0\text{ m/s}) = 2.0\text{ m/s}$$

The speed of the CM is $\boxed{2.0\text{ m/s}}$.

(b) Locate the center of mass between the disk and the center of the stick at impact:

$$y_{\text{CM}} = \frac{m_d r + m_s(0)}{m_d + m_s} = \frac{(2.0\text{ kg})(2.0\text{ m})}{2.0\text{ kg} + 1.0\text{ kg}} = \frac{4}{3}\text{ m}$$

This means at impact the CM is 4/3 meters from the center of the stick; therefore, the disk is 2.0 meters – 4/3 meters = 2/3 meters from the CM at impact. Use the parallel-axis theorem to find the moment of inertia of the system about the CM:

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$$I_s = I_{\text{CM}} + m_s r_s^2 = 1.33\ \text{kg}\cdot\text{m}^2 + (1.0\ \text{kg})\left(\frac{4}{3}\ \text{m}\right)^2 = 3.11\ \text{kg}\cdot\text{m}^2$$

The moment of inertia of the disk about the CM is

$$I_d = m_d r_d^2 = (2.0\ \text{kg})\left(\frac{2}{3}\ \text{m}\right)^2 = 0.889\ \text{kg}\cdot\text{m}^2$$

Angular momentum about the CM is conserved:

$$L = r_d m_d v_d = I_d\omega + I_s\omega = (I_d + I_s)\omega$$

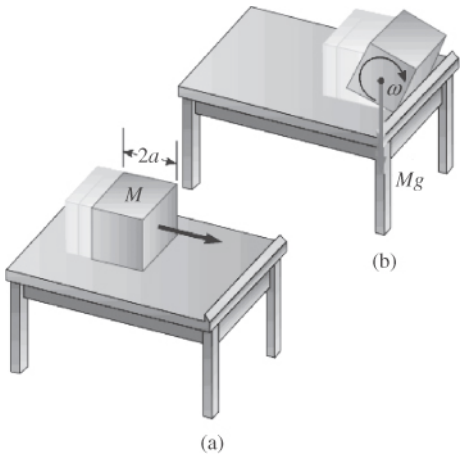
$$\omega = \frac{r_d m_d v_d}{I_d + I_s} = \frac{\left(\frac{2}{3}\ \text{m}\right)(2.0\ \text{kg})(3.0\ \text{m/s})}{0.889\ \text{kg}\cdot\text{m}^2 + 3.11\ \text{kg}\cdot\text{m}^2}$$

$$= \frac{4.0\ \text{kg}\cdot\text{m}^2/\text{s}}{4.00\ \text{kg}\cdot\text{m}^2} = \boxed{1.0\ \text{rad/s}}$$

P11.63 Angular momentum is conserved during the inelastic collision.

$$Mva = I\omega$$

$$\omega = \frac{Mva}{I} = \frac{3v}{8a}$$



ANS. FIG. P11.63

The condition, that the box falls off the table, is that the center of mass must reach its maximum height as the box rotates, $h_{\text{max}} = a\sqrt{2}$. Using conservation of energy:

$$\frac{1}{2}I\omega^2 = Mg\left(a\sqrt{2} - a\right)$$

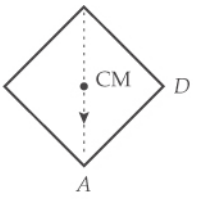
$$\frac{1}{2}\left(\frac{8Ma^2}{3}\right)\left(\frac{3v}{8a}\right)^2 = Mg\left(a\sqrt{2} - a\right)$$

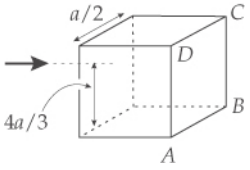
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$$v^2 = \frac{16}{3} ga\left(\sqrt{2} - 1\right)$$

$$v = \boxed{4\left[\frac{ga}{3}\left(\sqrt{2} - 1\right)\right]^{1/2}}$$

P11.64 For the cube to tip over, the center of mass (CM) must rise so that it is over the axis of rotation AB. To do this, the CM must be raised a distance of $a\left(\sqrt{2} - 1\right)$. After the bullet strikes the cube, the system is isolated:





ANS. FIG. P11.64

$$K_f + U_f = K_i + U_i$$

$$0 + Mga\left(\sqrt{2} - 1\right) = \frac{1}{2} I_{\text{cube}} \omega^2 + 0$$

The moment of inertia of the cube about its CM (from Table 10.2) is

$$I_{CM} = \frac{1}{12} M\left[(2a)^2 + (2a)^2\right] = \frac{8}{12} Ma^2 = \frac{2}{3} Ma^2$$

The cube rotates about an edge, $\sqrt{2}a$ from the CM. By the parallel-axis theorem,

$$I = I_{\text{CM}} + M\left(\sqrt{2}a\right)^2 = \frac{2}{3} Ma^2 + 2Ma^2 = \frac{8}{3} Ma^2$$

From conservation of angular momentum,

$$L_{i\ \text{(bullet)}} = L_{i\ \text{(cube)}} \rightarrow \frac{4a}{3} mv = \left(\frac{8}{3} Ma^2\right)\omega \quad \rightarrow \quad \omega = \frac{mv}{2Ma}$$

Inserting the expression for ω back into the energy equation, we have

$$Mga\left(\sqrt{2} - 1\right) = \frac{1}{2}\left(\frac{8}{3} Ma^2\right)\frac{m^2 v^2}{4M^2 a^2} \rightarrow v = \boxed{\frac{M}{m}\sqrt{3ga\left(\sqrt{2} - 1\right)}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P11.2 (a) 740 cm^2; (b) 59.5 cm

P11.4 See full solution in P11.4.

P11.6 (a) 168°; (b) 11.9°; (c) the first method

P11.8 (a) $(-10.0\ \text{N}\cdot\text{m})\hat{\mathbf{k}}$; (b) Yes; (c) Yes; (d) Yes; (e) No; (f) $5.00\hat{\mathbf{j}}$ m

P11.10 (a) No; (b) No, the cross product could not work out that way.

P11.12 $\left(-22.0\ \text{kg}\cdot\text{m}^2/\text{s}\right)\hat{\mathbf{k}}$

P11.14 (a) $\left(-9.03\times10^9\,\text{kg}\cdot\text{m}^2/\text{s}\right)\hat{\mathbf{j}}$; (b) No; (c) Zero

P11.16 $\sqrt{m^2 g\ell^3\dfrac{\sin^4\theta}{\cos\theta}}$

P11.18 (a) 3.14 N · m; (b) (0.480 kg · m)v; (c) 6.53 m/s^2

P11.20 (a) $2t^3\hat{\mathbf{i}}+t^2\hat{\mathbf{j}}$; (b) The particle starts from rest at the origin, starts moving into the first quadrant, and gains speed faster while turning to move more nearly parallel to the x axis; (c) $\left(12t\hat{\mathbf{i}}+2\hat{\mathbf{j}}\right)\ \text{m/s}^2$;

(d) $\left(60t\hat{\mathbf{i}}+10\hat{\mathbf{j}}\right)$ N; (e) $-40t^3\hat{\mathbf{k}}\ \text{N}\cdot\text{m}$;; (f) $-10t^4\hat{\mathbf{k}}\ \text{kg}\cdot\text{m}^2/\text{s}$;

(g) $(90t^4 + 10t^2)$ J; (h) $(360t^3 + 20t)$ W

P11.22 $\vec{\mathbf{L}}=\left(4.50\ \text{kg}\cdot\text{m}^2/\text{s}\right)\hat{\mathbf{k}}$

P11.24 $K=\dfrac{1}{2}I\omega^2=\dfrac{1}{2}\dfrac{I^2\omega^2}{I}=\dfrac{L^2}{2I}$

P11.26 (a) $7.06\times10^{33}\,\text{kg}\cdot\text{m}^2/\text{s}$, toward the north celestial pole;

(b) $2.66\times10^{40}\,\text{kg}\cdot\text{m}^2/\text{s}$, toward the north ecliptic pole; (c) See P11.26(c) for full explanation.

P11.28 8.63 m/s^2

P11.30 (a) $\dfrac{I_1}{I_1+I_2}\omega_i$; (b) $\dfrac{I_1}{I_1+I_2}$

P11.32 (a) 2.91 s; (b) Yes because there is no net external torque acting on the puck-rod-putty system; (c) No because the pivot pin is always pulling on the rod to change the direction of the momentum; (d) No. Some mechanical energy is converted into internal energy. The collision is perfectly inelastic.

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P11.34 (a) 1.91 rad/s; (b) 2.53 J, 6.44 J

P11.36 (a) $7.20\times10^{-3}\,\text{kg}\cdot\text{m}^2/\text{s}$; (b) 9.47 rad/s

P11.38 (a) 2.35 rad/s; (b) 0.498 rad/s; (c) 5.58°

P11.40 When the people move to the center, the angular speed of the station increases. This increases the effective gravity by 26%. Therefore, the ball will not take the same amount of time to drop.

P11.42 131 s

P11.44 (a) 0; (b) monkey and bananas move upward with the same speed; (c) The monkey will not reach the bananas.

P11.46 (a) $0.250\hat{\mathbf{i}}$ m/s; (b) 0.000 716; (c) $0.250\hat{\mathbf{i}}$ m/s; (d) 15.8 rad/s; (e) 1.00; (f) See P11.46(f) for full explanation.

P11.48 (a) 11.1 m/s; (b) $5.32\times10^{3}\,\text{kg}\cdot\text{m}^2/\text{s}$; (c) See P11.48(c) for full explanation; (d) 12.0 m/s; (e) 1.08 kJ

P11.50 (a) $2.11\hat{\mathbf{j}}$ rad/s; (b) See P11.50(b) for full problem statement; (c) Yes, with the left-hand side representing the final situation and the right-hand side representing the original situation, the equation describes the throwing process.

P11.52 (a) 4.50 m/s; (b) 10.1 N; (c) 0.450 J

P11.54 An asteroid that would cause a 0.500-s change in the rotation period of the Earth has a mass of 1.38×10^{19} kg and is an order of magnitude larger in diameter than the one that caused the extinction of the dinosaurs.

P11.56 (a) Mvd; (b) Mv^2; (c) Mvd; (d) $2v$; (e) $4Mv^2$; (f) $3Mv^2$

P11.58 (a) $\omega_f = \dfrac{36.0(1+3.20m)}{1+20.0m}$ rad/s; (b) ω_f decreases smoothly from a maximum value of 36.0 rad/s for $m = 0$ toward a minimum value of $(36 \times 3.2/20) = 5.76$ rad/s as $m \to \infty$

P11.60 5.99×10^{-2} J

P11.62 (a) 2.0 m/s; (b) 1.0 rad/s

P11.64 $\dfrac{M}{m}\sqrt{3ga\left(\sqrt{2}-1\right)}$

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